

PRIMARY 3 MATHEMATICS SCHEME OF WORK

FIRST TERM

Week	Topic	Content Breakdown
1	Whole Numbers up to 999	Reading and writing numbers up to 999; place value (hundreds, tens, and ones).
2	Whole Numbers up to 999	Comparing and ordering numbers; use of symbols ($<$, $>$, $=$) and number patterns.
3	Whole Numbers up to 999	Rounding numbers to the nearest ten and hundred; number sequences.
4	Fractions	Introduction to fractions as parts of a whole ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$). Identifying fractions in everyday objects.
5	Midterm Test	—
6	Fractions	Comparing and arranging simple fractions; addition and subtraction of fractions with the same denominator.
7	Midterm Break	—
8	Fractions	Word problems involving fractions in real-life situations.
9	Fractions	Fraction representation on number lines and shapes.
10	Revision	Review of whole numbers and fractions.
11–13	Examination / Closing	—

PRIMARY 3 MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: WHOLE NUMBERS UP TO 999.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Read and write numbers up to 999 in figures and words.
- Identify the place value of digits in a 3-digit number (Hundreds, Tens, and Ones).
- Expand 3-digit numbers using their place values.

CONTENT Reading and Writing Numbers up to 999: Numbers are like a special code made from digits.

- **Digits** are the ten symbols we use: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- When we put one digit alone, we get a **one-digit number** like 5 or 9. It tells us how many single items there are (ones).
- When we put two digits together, we get a **two-digit number** like 24 or 50. The first digit tells us how many groups of ten we have, and the last digit tells us the extra ones.
- When we put three digits together, we get a **three-digit number** like 128 or 999. These numbers can go as high as 999.

To write a number in words, we read it from left to right.

1. Look at the hundreds digit first.
2. Then look at the last two digits together.

Example:

- **Figure:** 245
- The first digit (2) is in the hundreds place, so we say "Two hundred".
- The last two digits (45) are "forty-five".
- **Words:** Two hundred and forty-five. (We use "and" to connect the hundreds to the rest).

Figure: 703

- **Hundreds:** Seven hundred.
- **Tens and Ones:** The digit in the tens place is 0, which means no tens. We only have 3 ones.
- **Words:** Seven hundred and three.



Place Value (Hundreds, Tens, and Ones): Every digit in a number has its own special "home" or place.

- The place tells us the real value of that digit.
- For 3-digit numbers, we use three columns: **Hundreds (H)**, **Tens (T)**, **Ones (O)**.

PLACE VALUE		
H	T	O
3	4	5

Example: 345

- The digit **3** is in the **Hundreds** place.
- Its **value is 300** (which is 3 groups of one hundred).
- The digit **4** is in the **Tens** place.
- Its **value is 40** (which is 4 groups of ten).
- The digit **5** is in the **Ones** place.
- Its **value is 5** (which is 5 single items).

Expanding Numbers: Expanding a number means "stretching it out" to show the real value of each digit. We add the values of the hundreds, tens, and ones together.

- $672 = 6 \text{ Hundreds} + 7 \text{ Tens} + 2 \text{ Ones}$
- **Expansion:** $600 + 70 + 2$
- $908 = 9 \text{ Hundreds} + 0 \text{ Tens} + 8 \text{ Ones}$
- **Expansion:** $900 + 0 + 8 = 900 + 8$

EVALUATION

1. Write the number **456** in words.
2. Write "**Seven hundred and twenty-one**" using digits (figures).
3. In the number **982**, what is the place value of the digit **8**? (Is it Hundreds, Tens, or Ones?)
4. Expand the number **135** by writing it as an addition sentence (e.g., $__ + __ + __$).
5. What is the actual value of the digit **5** in the number **500**?
6. Using the digits 3, 1, and 9 only once, arrange them to form the smallest possible 3-digit number.

ASSIGNMENT

- **(Written):** Write these numbers in words: (a) 315 (b) 890 (c) 105.
- **(Drawing):** Draw a Place Value chart. Write the number **674** in it, placing each digit in its correct column (H, T, O).

- **(Practical/Oral):** Find three small objects (like pebbles or seeds). Group them into 9 groups of ten and have 9 left over. Tell your parent what number you have made (99). Then add one more object and say the new number (100).
-



WEEK 2: WHOLE NUMBERS UP TO 999.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Compare two numbers using the symbols $<$, $>$, and $=$.
- Order a set of numbers in ascending (small to big) order.
- Order a set of numbers in descending (big to small) order.

CONTENT Comparing Numbers: We compare numbers to see which is bigger (greater) or smaller (less). We use three special signs:

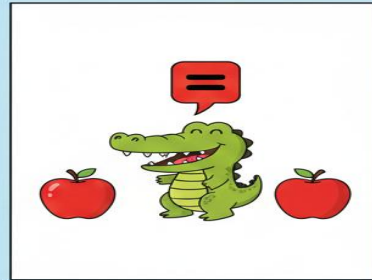
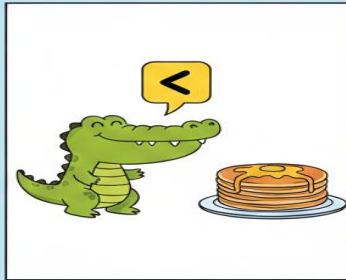
- **$>$ means Greater Than (Bigger):** Think of it as a hungry crocodile's open mouth that always wants to eat the bigger number!
- Example: $9 > 3$ (Nine is greater than three).
- **$<$ means Less Than (Smaller):** It's like the crocodile's mouth is closed because the number is small.
- Example: $3 < 9$ (Three is less than nine).
- **$=$ means Equal To (The same amount).**
- Example: $5 = 5$ (Five is equal to five).

How to Compare 3-Digit Numbers (Step-by-Step):

1. **First, look at the Hundreds place:** The number with the bigger hundreds digit is the bigger number.
 - Example: Compare 420 and 220.
 - 4 hundreds is more than 2 hundreds. So, $420 > 220$.
2. **If the Hundreds are the same, look at the Tens place.**
 - Example: Compare 560 and 580.
 - Hundreds are the same (5). 8 tens is more than 6 tens. So, $560 < 580$.
3. **If the Hundreds and Tens are the same, look at the Ones place.**
 - Example: Compare 342 and 342.

- Hundreds (3) and Tens (4) are the same. Ones (2) are the same. So, $342 = 342$.

COMPARING NUMBERS



**THE GREATER GATOR
ALWAYS EATS THE BIGGER NUMBER!**

Ordering Numbers

- **Ascending Order:** Arranging numbers from the **smallest** to the **biggest**.
- Imagine walking **up** the stairs, starting from the lowest step.
- Example: 10, 50, 200, 500.
- **Descending Order:** Arranging numbers from the **biggest** to the **smallest**.
- Imagine walking **down** the stairs, starting from the highest step.
- Example: 500, 200, 50, 10.

Number Patterns (Skip Counting): A pattern is a sequence that follows a rule. A simple rule is to skip count by a certain number.

- **Skip counting in 2s:** 2, 4, 6, 8, 10... (Like counting pairs of shoes).
- **Skip counting in 10s:** 10, 20, 30, 40... (Like counting bundles of ten sticks).

- **Skip counting in 100s:** 100, 200, 300, 400... (Like counting bags of 100 cowries).

EVALUATION

1. Put the correct sign ($>$ or $<$) between: 500 ____ 300.
2. Put the correct sign ($>$ or $<$) between: 120 ____ 190.
3. Arrange these numbers in ascending order (smallest first): 340, 100, 800, 250.
4. Arrange these numbers in descending order (biggest first): 55, 999, 105, 400.
5. Complete the pattern: 200, 300, 400, ____, ____.
6. Which number is smaller: 718 or 781? Explain how you know by looking at the tens digits.

ASSIGNMENT

- **(Written):** Use $<$, $>$, or $=$ to complete: (a) 909 ____ 990 (b) 450 ____ 450.
 - **(Written):** Write these numbers from smallest to biggest: 600, 200, 900, 100.
 - **(Practical/Oral):** Clap and count in 5s up to 30: 5, 10, 15, ____, ____, 30. Write down the two missing numbers.
-

WEEK 3: WHOLE NUMBERS UP TO 999.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Round numbers to the nearest Ten.
- Round numbers to the nearest Hundred.
- Complete simple number sequences.

CONTENT Rounding Numbers to the Nearest Ten: Rounding helps us estimate or find a number that is easier to work with. When rounding to the nearest ten, we want to find the closest multiple of 10 (like 10, 20, 30, etc.). We look at the **Ones** digit.

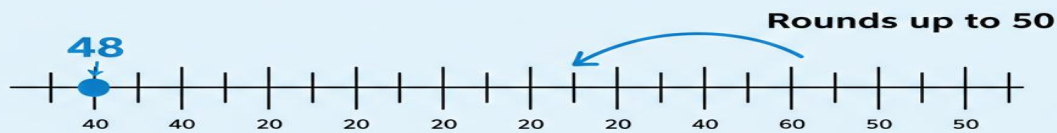
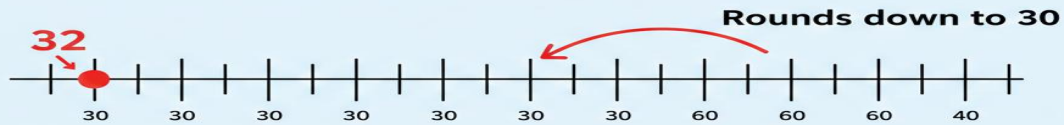
The Rule:

- **If the Ones digit is 0, 1, 2, 3, or 4:** We round **DOWN** to the ten that is already there. The Tens digit stays the same, and the Ones become 0.
- **If the Ones digit is 5, 6, 7, 8, or 9:** We round **UP** to the next ten. We add 1 to the Tens digit, and the Ones become 0.

Examples:

- **32:** The Ones digit is 2. (2 is 0-4). Round DOWN. 32 is between 30 and 40, but closer to **30**.
- **48:** The Ones digit is 8. (8 is 5-9). Round UP. 48 is between 40 and 50, but closer to **50**.
- **155:** The Ones digit is 5. (5 is 5-9). Round UP. 155 is between 150 and 160. Adding 1 to the tens (5 tens becomes 6 tens) gives us **160**.

NUMBER ROUNDING



Rounding Numbers to the Nearest Hundred: When rounding to the nearest hundred, we find the closest multiple of 100 (like 100, 200, 300). We look at the **Tens** digit.

The Rule:

- **If the Tens digit is 0, 1, 2, 3, or 4:** Round **DOWN**. The Hundreds digit stays the same, and the Tens and Ones become 0.
- **If the Tens digit is 5, 6, 7, 8, or 9:** Round **UP**. Add 1 to the Hundreds digit, and the Tens and Ones become 0.

Examples:

- **320:** The Tens digit is 2. (2 is 0-4). Round **DOWN**. 320 is between 300 and 400, closer to **300**.
- **680:** The Tens digit is 8. (8 is 5-9). Round **UP**. 680 is between 600 and 700, closer to **700**.

Number Sequences: A number sequence is a pattern of numbers. To find the next number, we must first discover the rule. The rule could be "add the same number each time" or "subtract the same number each time."

- **Sequence:** 10, 20, 30, ____?

- **Rule:** Add 10 each time. $30 + 10 = 40$.
- **Sequence:** 50, 45, 40, ____?
- **Rule:** Subtract 5 each time. $40 - 5 = 35$.

EVALUATION

1. Round 43 to the nearest ten.
2. Round 67 to the nearest ten.
3. Round 125 to the nearest hundred. (Hint: Look at the tens digit).
4. Round 890 to the nearest hundred.
5. What is the next number in this pattern: 2, 4, 6, 8, ____?
6. Complete the sequence: 100, 90, 80, ____.

ASSIGNMENT

- **(Written):** Round to the nearest Ten: (a) 24 (b) 79 (c) 15.
 - **(Written):** Round to the nearest Hundred: (a) 230 (b) 560 (c) 850.
 - **(Practical/Oral):** With a family member, count backwards in 50s from 300: 300, 250, 200, ____, ____. Write down the next two numbers.
-

WEEK 4: FRACTIONS.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define a fraction as a part of a whole.
- Identify and interpret the fractions $\frac{1}{2}$ (half), $\frac{1}{4}$ (quarter), and $\frac{3}{4}$ (three-quarters).
- Identify the numerator and denominator in a fraction.

CONTENT What is a Fraction?: A fraction is a part of a whole thing.

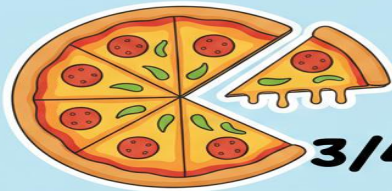
- When you break or share something into equal pieces, each piece is a fraction of the whole.
- Imagine one whole orange. If you cut it into two equal pieces, each piece is a **half** of the orange.
- If you cut a loaf of bread into 4 equal slices for your family, each slice is a **quarter** of the bread.

Parts of a Fraction: A fraction is written as two numbers, one on top of the other, with a line between them.

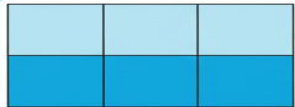
UNDERSTANDING FRACTIONS

Numerator → $\frac{3}{4}$ ← **Denomimor**

$\frac{3}{4}$



$\frac{3}{4}$



$\frac{3}{4}$

**The Numerator is the number of parts we have.
The Denomintor is the total nubor of equal parts.**

- **Top Number (Numerator):** This tells us **how many parts** we have or are talking about. (e.g., how many slices you ate).
- **Bottom Number (Denominator):** This tells us **into how many equal parts** the whole thing was divided. (e.g., how many slices the whole pizza was cut into).

Common Fractions

- **One Half ($1/2$):** The whole is divided into **2 equal parts**. One of those parts is "one half".
- Example: Cutting an apple into two equal pieces. Folding a paper perfectly into two matching sides.
- **One Quarter ($1/4$):** The whole is divided into **4 equal parts**. One of those parts is "one quarter".
- Example: Cutting a square of *moin-moin* into 4 small equal squares.
- **Three Quarters ($3/4$):** The whole is divided into **4 equal parts**. We are talking about **3** of those parts.
- Example: If you eat 3 slices of a pizza cut into 4 slices, you have eaten three-quarters.

Fractions in Everyday Nigerian Life

- **Time:** "Half past 7" means the minute hand has moved halfway around the clock.
- **Market:** "A quarter of a *mudu* of rice" means one part out of four equal parts of a *mudu* measure.
- **Food:** Sharing a *chin-chin* pack equally among 4 friends means each gets one-quarter ($1/4$) of the pack.

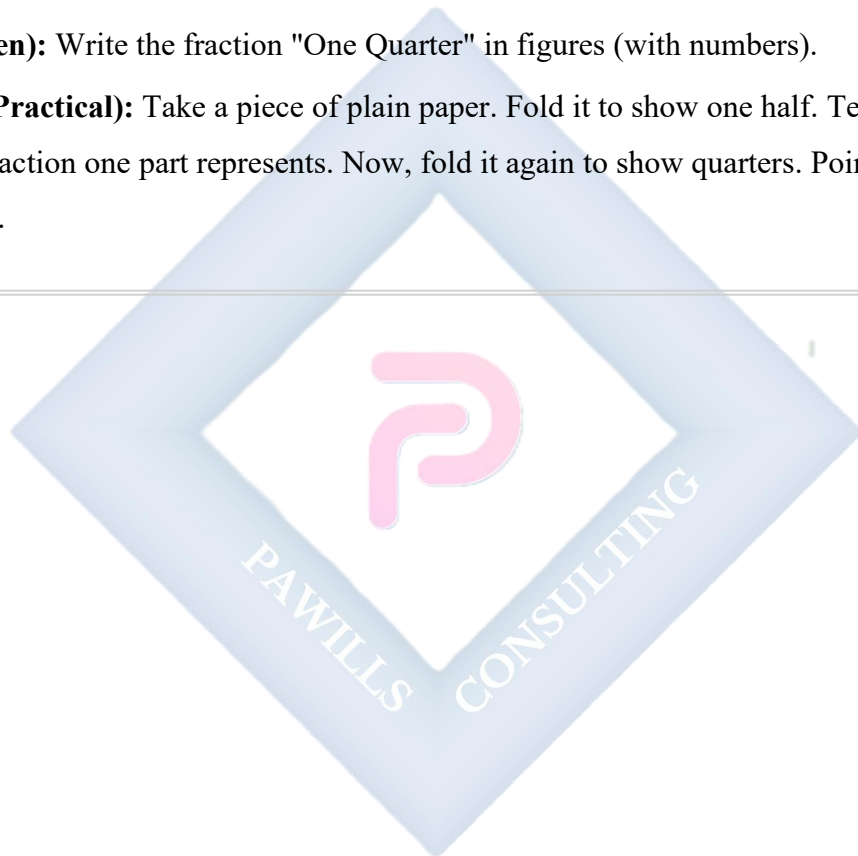
EVALUATION

1. In a fraction, what does the bottom number (denominator) tell us?
2. Draw a circle and shade **one-half ($1/2$)** of it.
3. Draw a square, divide it into 4 equal parts, and shade **one-quarter ($1/4$)**.

4. If a *suya* stick is cut into 4 equal pieces and you eat 1 piece, what fraction of the *suya* did you eat?
5. Write the fraction for "one half" using numbers.
6. In the fraction $\frac{3}{4}$, which number is the numerator?

ASSIGNMENT

- **(Drawing):** Draw a rectangle. Divide it into 4 equal parts (columns or rows). Shade 3 parts. Write the fraction for the shaded part.
- **(Written):** Write the fraction "One Quarter" in figures (with numbers).
- **(Oral/Practical):** Take a piece of plain paper. Fold it to show one half. Tell your parent what fraction one part represents. Now, fold it again to show quarters. Point to one quarter.



WEEK 5: MIDTERM TEST.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Attempt questions based on topics covered from Week 1 to Week 4.

CONTENT Assessment Preparation: Today, you will show what you have learned so far. The test will have different types of questions. Remember to:

- Read each question carefully.
- Take your time and think.
- Check your work when you are done.
- Write neatly so your teacher can read your answers.

The test will cover:

- **Reading and writing numbers** up to 999 in words and figures.
- **Place value** of digits in hundreds, tens, and ones places.
- **Expanding numbers** (e.g., $234 = 200 + 30 + 4$).
- **Comparing numbers** using $>$, $<$, and $=$.
- **Ordering numbers** from smallest to biggest (ascending) and biggest to smallest (descending).
- **Rounding numbers** to the nearest ten and hundred.
- **Fractions:** Identifying $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$, and knowing the numerator and denominator.

EVALUATION

1. The teacher will administer the written test.
2. After the test, the teacher will guide the class in marking and correcting the answers together.

ASSIGNMENT

- **(Written):** Correct any three mistakes you made on your test paper. Write the correct answer neatly next to it.

- **(Oral):** Explain to someone at home one thing you found easy on the test and one thing you found tricky.
 - **(Practical):** Choose any topic from Weeks 1-4 (like place value or fractions) and teach a younger sibling or a toy how to do one simple activity from it.
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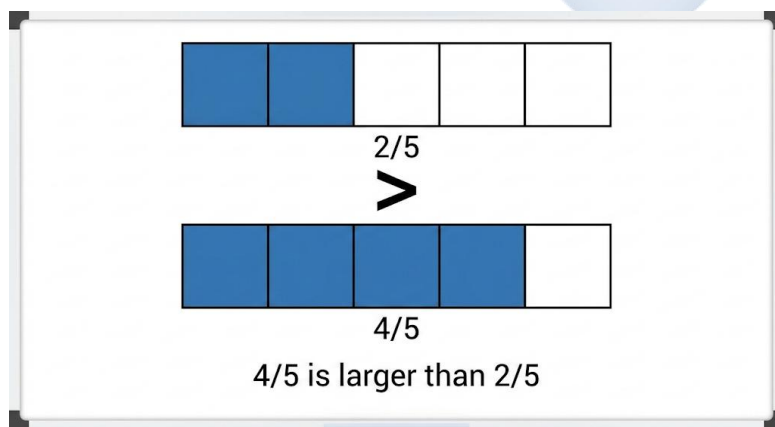
WEEK 6: FRACTIONS.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Compare simple fractions with the same denominator.
- Add two fractions with the same denominator.
- Subtract two fractions with the same denominator.

CONTENT Comparing Fractions (Same Denominator): When fractions have the same bottom number (denominator), it means the whole was cut into the same size pieces. To compare, we just look at the top number (numerator). **More pieces means a bigger fraction.**

- **Example 1:** Compare $\frac{1}{4}$ and $\frac{3}{4}$.
- Imagine one whole *agege* bread cut into 4 equal slices.
- 1 slice ($\frac{1}{4}$) is less than 3 slices ($\frac{3}{4}$).
- So, $\frac{3}{4} > \frac{1}{4}$.
- **Example 2:** Compare $\frac{2}{5}$ and $\frac{4}{5}$.
- A rope is divided into 5 equal parts.
- 2 parts is less than 4 parts.
- So, $\frac{4}{5} > \frac{2}{5}$.



Addition of Fractions (Same Denominator): When adding fractions with the same denominator, we are putting together pieces of the same size. We **only add the top numbers (numerators)**. The bottom number (denominator) stays the same because the size of the pieces doesn't change.

- **Rule:** Add the numerators, keep the denominator.
- **Example:** $\frac{1}{4} + \frac{2}{4} = ?$
- **Step 1:** Add the top numbers: $1 + 2 = 3$.
- **Step 2:** Keep the bottom number: **4**.
- **Answer:** $\frac{3}{4}$. (1 slice plus 2 slices equals 3 slices out of 4).

Subtraction of Fractions (Same Denominator): When subtracting fractions with the same denominator, we are taking away some pieces from a total. We **only subtract the top numbers (numerators)**. The bottom number (denominator) stays the same.

- **Rule:** Subtract the numerators, keep the denominator.
- **Example:** $\frac{3}{5} - \frac{1}{5} = ?$
- **Step 1:** Subtract the top numbers: $3 - 1 = 2$.
- **Step 2:** Keep the bottom number: **5**.
- **Answer:** $\frac{2}{5}$. (If you had 3 slices out of 5 and ate 1, you have 2 left).

EVALUATION

1. Which is bigger: $\frac{2}{6}$ or $\frac{5}{6}$?
2. Which is smaller: $\frac{1}{3}$ or $\frac{2}{3}$?
3. Add: $\frac{2}{7} + \frac{3}{7}$.
4. Add: $\frac{1}{5} + \frac{1}{5}$.
5. Subtract: $\frac{4}{8} - \frac{3}{8}$.
6. Chinedu had $\frac{6}{10}$ of a notebook used. He tore out $\frac{2}{10}$ for a test. What fraction of the notebook is left? (Write the subtraction sentence).

ASSIGNMENT

- **(Written):** Arrange these fractions from smallest to biggest: $\frac{3}{9}$, $\frac{1}{9}$, $\frac{7}{9}$.

- **(Written):** Solve: $\frac{3}{8} + \frac{2}{8} = \underline{\hspace{1cm}}$.
 - **(Practical/Drawing):** Draw a circle and divide it into 10 equal parts (like a clock). Shade 7 parts to show $\frac{7}{10}$. Then cross out 4 shaded parts to show subtraction. Write the answer:
 $\frac{7}{10} - \frac{4}{10} = \underline{\hspace{1cm}}$.
-



WEEK 7: MIDTERM BREAK.



WEEK 8: FRACTIONS.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify fractions in word problems.
- Solve simple word problems involving addition of fractions.
- Solve simple word problems involving subtraction of fractions.

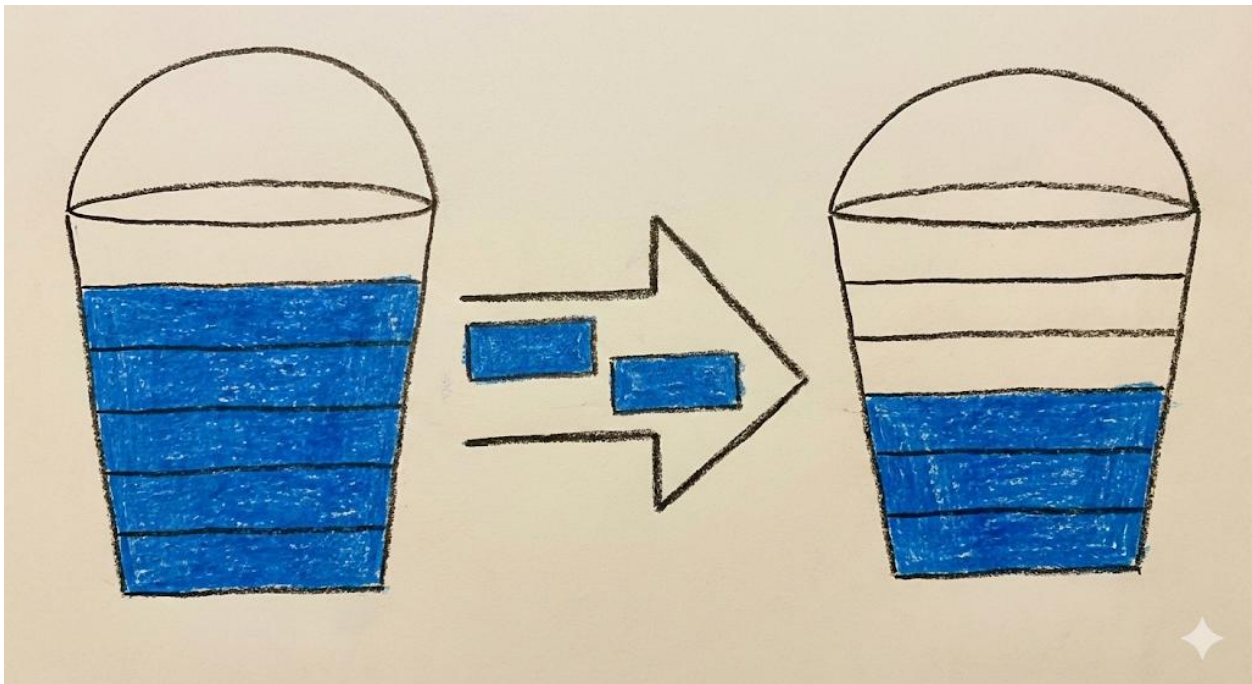
CONTENT Word Problems: Addition: To solve word problems, listen or read carefully for clues. Words like "**altogether**," "**total**," "**sum**," "**both**," "**in all**" tell us to add.

- **Example 1:**
- **Problem:** Amina drank $\frac{1}{5}$ of a bottle of water in the morning. She drank another $\frac{2}{5}$ in the afternoon. What fraction of the water did she drink altogether?
- **Understanding:** "Altogether" means we add. The water bottle is the whole, cut into 5 equal parts (fifths).
- **Solution:**
- Morning: $\frac{1}{5}$
- Afternoon: $\frac{2}{5}$
- Total = $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$.
- **Answer:** Amina drank $\frac{3}{5}$ of the water.

Word Problems: Subtraction: Look for words like "**left**," "**remaining**," "**how much more**," "**difference**," "**gave away**". These tell us to subtract.

- **Example 2:**
- **Problem:** A bucket was $\frac{5}{6}$ full of water. Mama used $\frac{2}{6}$ of the water to cook. What fraction of water is left in the bucket?
- **Understanding:** "Left" means we subtract. The bucket is the whole, in sixths.

- **Solution:**
- Start: $\frac{5}{6}$ full.
- Used: $\frac{2}{6}$.
- Left = $\frac{5}{6} - \frac{2}{6} = \frac{3}{6}$.
- **Answer:** $\frac{3}{6}$ of the water is left.



Real-life Nigerian Situations

- **Money:** If you spend $\frac{1}{4}$ of your ₦20 on *gala* and another $\frac{1}{4}$ on *pure water*, you have spent $\frac{2}{4}$ (or $\frac{1}{2}$) of your money.
- **Farming:** If a farmer plants maize on $\frac{3}{8}$ of his plot and beans on $\frac{4}{8}$, he has used $\frac{7}{8}$ of his plot for farming.

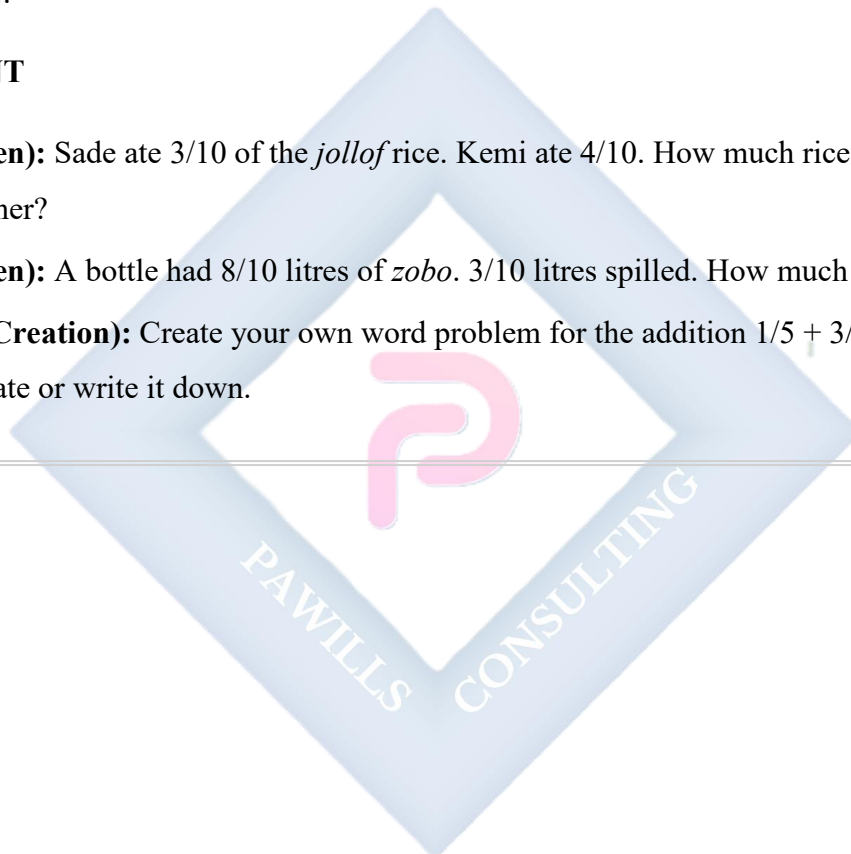
EVALUATION

1. Tunde has $\frac{2}{7}$ of a loaf of bread. His friend gives him $\frac{3}{7}$ more. How much bread does Tunde have now?

2. A water tank is $\frac{7}{10}$ full. The family used $\frac{4}{10}$ of the water for washing. What fraction of water is remaining?
3. Ngozi read $\frac{1}{3}$ of a storybook on Monday and $\frac{1}{3}$ on Tuesday. What total fraction of the book did she read?
4. A whole orange can be written as $\frac{4}{4}$. If you eat $\frac{1}{4}$ of it, what fraction of the orange is left?
5. Mr. Bala painted $\frac{2}{9}$ of his fence red and $\frac{5}{9}$ of it blue. What fraction of the fence is painted?

ASSIGNMENT

- **(Written):** Sade ate $\frac{3}{10}$ of the *jollof* rice. Kemi ate $\frac{4}{10}$. How much rice did they eat altogether?
- **(Written):** A bottle had $\frac{8}{10}$ litres of *zobo*. $\frac{3}{10}$ litres spilled. How much *zobo* is left?
- **(Oral/Creation):** Create your own word problem for the addition $\frac{1}{5} + \frac{3}{5}$. Tell it to a classmate or write it down.



WEEK 9: FRACTIONS.

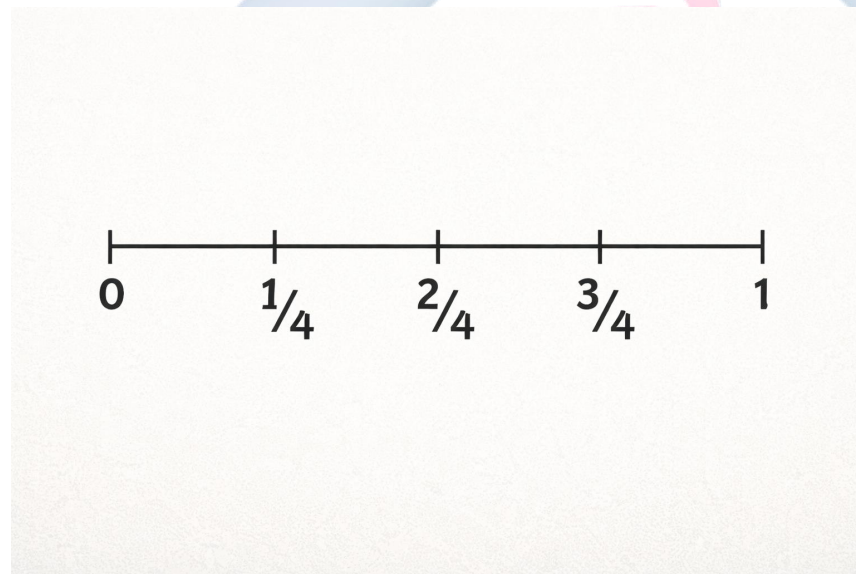
OBJECTIVES By the end of the lesson, pupils should be able to:

- Represent fractions on a number line.
- Represent fractions using shapes and shading.
- Identify a fraction by looking at a shaded diagram.

CONTENT Fractions on a Number Line: A number line isn't just for whole numbers. We can show fractions between 0 and 1. The space from 0 to 1 represents one whole thing.

How to show quarters ($\frac{1}{4}$, $\frac{2}{4}$, $\frac{3}{4}$):

1. Draw a straight line. Mark 0 on the left end and 1 on the right end.
2. Divide the distance between 0 and 1 into **4 equal jumps** or parts.
3. The first mark from 0 is $\frac{1}{4}$. The second mark is $\frac{2}{4}$ (which is the same as $\frac{1}{2}$). The third mark is $\frac{3}{4}$.



Fractions using Shapes (Circles, Squares, Rectangles): We can use any shape to model a fraction. **Steps:**

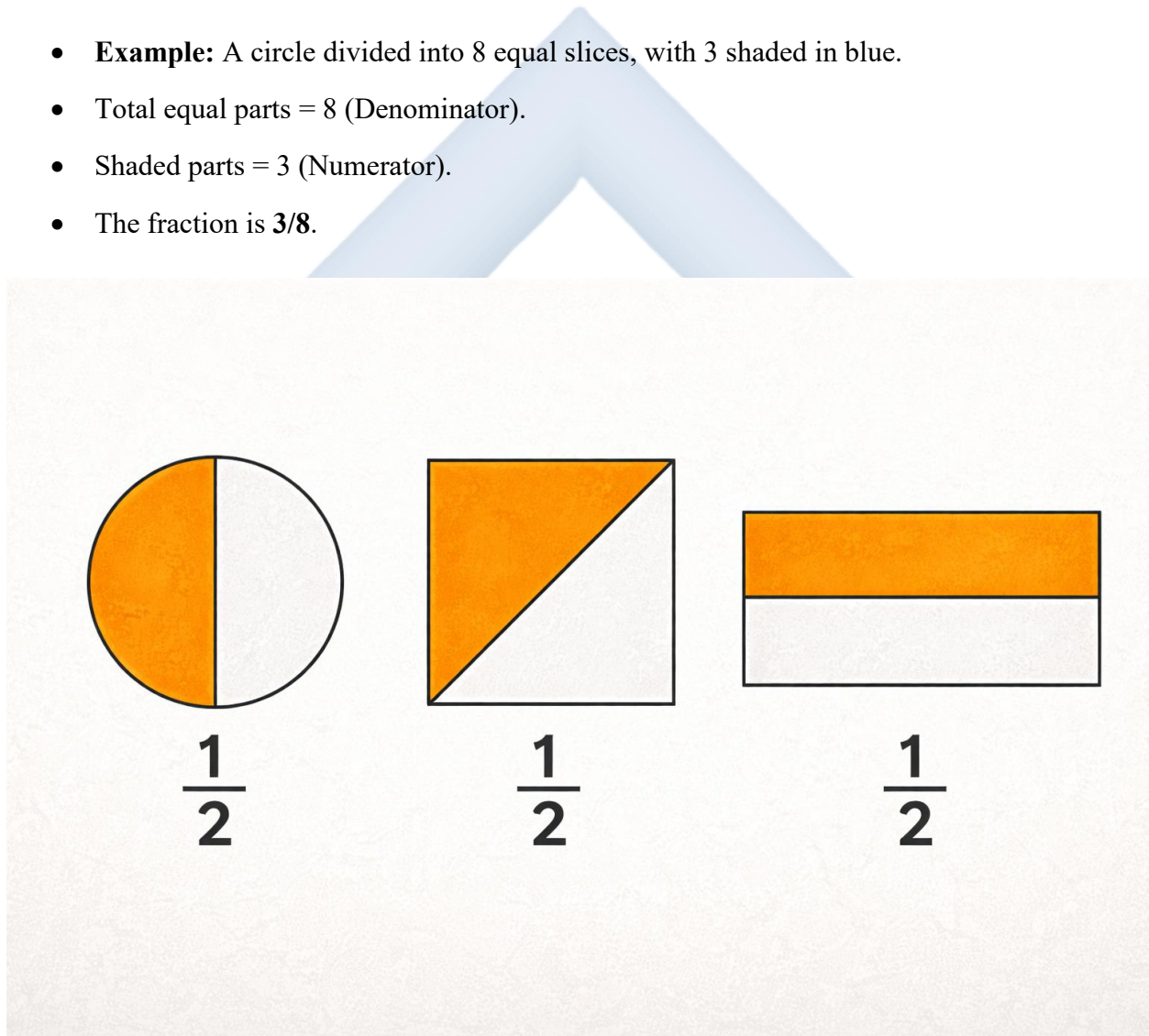
1. Draw a shape (the whole).
2. Divide it into equal parts. The number of parts is the **denominator**.

3. Shade some of the parts. The number of shaded parts is the **numerator**.

- **Example:** Represent $\frac{2}{3}$ with a rectangle.
- Draw a rectangle.
- Split it into **3** equal parts (strips or columns).
- Shade **2** of those parts.

Interpreting Diagrams (Finding the Fraction): If you see a picture of a divided shape, you can find the fraction.

- **Example:** A circle divided into 8 equal slices, with 3 shaded in blue.
- Total equal parts = 8 (Denominator).
- Shaded parts = 3 (Numerator).
- The fraction is $\frac{3}{8}$.



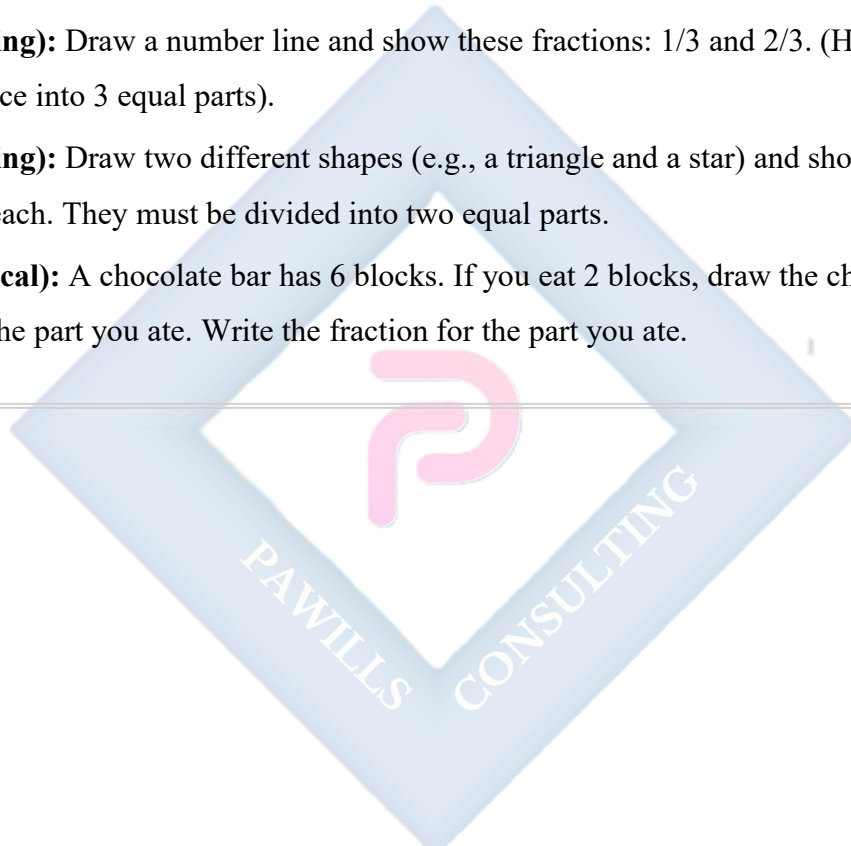
EVALUATION

1. Draw a number line from 0 to 1. Mark and label the point that shows **one-half** ($\frac{1}{2}$).

2. Draw a circle. Divide it into 4 equal parts and shade **three-quarters ($\frac{3}{4}$)**.
3. Look at this shape: [Teacher draws a square divided into 4 smaller squares, with 1 shaded]. What fraction is shaded?
4. Draw a rectangle to represent the fraction **$\frac{5}{8}$** .
5. To show the fraction $\frac{1}{5}$ on a number line, into how many equal parts must I divide the space between 0 and 1?

ASSIGNMENT

- **(Drawing):** Draw a number line and show these fractions: $\frac{1}{3}$ and $\frac{2}{3}$. (Hint: Divide the 0-1 space into 3 equal parts).
- **(Drawing):** Draw two different shapes (e.g., a triangle and a star) and show the fraction $\frac{1}{2}$ in each. They must be divided into two equal parts.
- **(Practical):** A chocolate bar has 6 blocks. If you eat 2 blocks, draw the chocolate bar and shade the part you ate. Write the fraction for the part you ate.



WEEK 10: REVISION.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Recall and answer questions on whole numbers (Place value, ordering, rounding).
- Recall and answer questions on fractions (Identification, operations, word problems).

CONTENT Review of Whole Numbers (Weeks 1-3): Let's remember the key ideas:

- **Place Value:** Hundreds, Tens, Ones. The digit's place tells its value (e.g., '7' in 572 is 70).
- **Expanding:** $841 = 800 + 40 + 1$.
- **Comparing & Ordering:** Use $>$, $<$, $=$. For ascending order, start with the smallest.
- **Rounding:**
 - To nearest ten: Look at the ones digit (0-4 down, 5-9 up).
 - To nearest hundred: Look at the tens digit (0-4 down, 5-9 up).

Review of Fractions (Weeks 4, 6, 8, 9): Let's remember the key ideas:

- **Meaning:** Part of a whole. Numerator (parts you have), Denominator (total equal parts).
- **Comparing:** Same denominator? Compare numerators.
- **Adding/Subtracting:** Same denominator? Add/subtract numerators, keep denominator.
- **Word Problems:** Find the clue words (altogether = add, left = subtract).
- **Showing Fractions:** On number lines and by shading shapes.

EVALUATION The teacher will write mixed questions on the board. Solve them carefully.

1. Write **815** in words.
2. Expand **452**.
3. Round **36** to the nearest ten.
4. Add: $\frac{2}{9} + \frac{5}{9}$.
5. Draw a square and shade **one-quarter** ($\frac{1}{4}$) of it.

ASSIGNMENT

- **(Written):** Solve: $346 + 128$. Then, round your answer to the nearest hundred.

- **(Written):** A cake was cut into 8 equal slices. If 3 slices are eaten, what fraction of the cake is left? Write the subtraction sentence and answer.
 - **(Oral Review):** Ask a family member to quiz you on 5 multiplication facts from the 2, 3, or 4 times table. (e.g., "What is 4×3 ?").
-



PRIMARY 3 MATHEMATICS SCHEME OF WORK

SECOND TERM

Week	Topic	Content Breakdown
1	Addition	Addition of 2-digit and 3-digit numbers without carrying.
2	Addition	Addition of 2-digit and 3-digit numbers with carrying; solving addition word problems.
3	Subtraction	Subtraction of 2-digit and 3-digit numbers without borrowing.
4	Subtraction	Subtraction with borrowing; solving subtraction word problems.
5	Midterm Test	—
6	Multiplication	Multiplication as repeated addition; times tables ($2\times$, $3\times$, $4\times$, $5\times$).
7	Midterm Break	—
8	Multiplication	Multiplying 2-digit numbers by 1-digit numbers; solving multiplication word problems.
9	Division	Division as sharing equally; relationship between multiplication and division.
10	Division	Solving simple division problems with remainders.
11–13	Revision / Examination / Closing	—

PRIMARY 3 MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: ADDITION (WITHOUT CARRYING).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Add 2-digit numbers without carrying.
- Add 3-digit numbers without carrying.
- Arrange numbers vertically according to their place value (Hundreds, Tens, and Ones) before adding.

CONTENT Understanding Addition: Addition is putting things together to find a total.

- It is like gathering all your toys into one box.
- When we add numbers on paper, we must be very tidy and line them up correctly so we add the right places together.
- We always start adding from the right side—the **Ones** place.

Arranging Numbers Vertically: To add correctly, numbers must be stacked in columns under their correct place value headings: **H**undreds (H), **T**ens (T), and **O**nes (O). This keeps tens with tens and ones with ones.

H	T	O

Each letter has its own column

Worked Example 1: Adding 2-Digit Numbers ($45 + 23$)

1. **Step 1: Arrange in columns.** Write the numbers so the tens and ones line up.

T O

4 5

2 3

2. **Step 2: Add the Ones column (O).** $5 + 3 = 8$. Write 8 under the Ones column.
3. **Step 3: Add the Tens column (T).** $4 + 2 = 6$. Write 6 under the Tens column.
- **Answer:** 68.

Worked Example 2: Adding 3-Digit Numbers (214 + 352)

1. **Step 1: Arrange in columns.** Make sure hundreds, tens, and ones are lined up.

H T O

2 1 4

3 5 2

2. **Step 2: Add Ones.** $4 + 2 = 6$. Write 6 under O.
3. **Step 3: Add Tens.** $1 + 5 = 6$. Write 6 under T.
4. **Step 4: Add Hundreds.** $2 + 3 = 5$. Write 5 under H.
- **Answer:** 566.

Hundreds	Tens	Ones
2	3	5
3	3	1
5	6	6

566
Final Answer: 566

EVALUATION

1. Add $24 + 12$.
2. Add $60 + 30$.
3. Arrange 521 and 143 vertically and find their sum.
4. Find the sum of 305 and 242.
5. $111 + 222 = ?$
6. Add 43 and 5. (Remember to line up the ones: $43 + 05$).

ASSIGNMENT

- **(Written):** Calculate: $72 + 16$.
- **(Written):** Calculate: $453 + 215$.
- **(Practical/Oral):** A boy has 12 marbles. His sister has 15 marbles. They put them all in one bag. How many marbles are in the bag? Draw the marbles or use seeds to act it out, then write the addition sentence.

WEEK 2: ADDITION (WITH CARRYING AND WORD PROBLEMS).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Add numbers involving "carrying" (renaming).
- Solve simple word problems involving addition.
- Identify keywords in word problems like "total," "sum," and "altogether."

CONTENT Addition with Carrying: Sometimes when we add a column, the answer is 10 or more. We can only write one digit in a column. The extra digit is called the "carry over" or "remainder." We carry it to the next column on the left (the Tens column).

Worked Example 1: $58 + 24$

1. **Step 1: Arrange.**

T O

5 8

2 4

2. **Step 2: Add Ones.** $8 + 4 = 12$.

- 12 means 1 Ten and 2 Ones.
- Write **2** in the Ones column.
- **Carry** the **1** (Ten) over to the top of the Tens column.
- Write a small '1' above the 5.

3. **Step 3: Add Tens.** Don't forget the carried number! Add: 1 (carried) $+ 5 + 2 = 8$.

- Write 8 in the Tens column.
- **Answer:** 82.

58 + 24

Tens	Ones
8 4	5 2
8 + 4	2 1
12	8
2	

82

Final Answer: 82

Solving Word Problems: To know when to add in a story problem, listen for special words: "total," "sum," "altogether," "in all," "both combined."

Worked Example 2 (Word Problem): Mr. Okon sold 125 yams on Monday and 148 yams on Tuesday. What is the total number of yams sold?

1. **Step 1:** Find the numbers to add: 125 and 148. The word "total" tells us to add.
2. **Step 2:** Arrange and add with carrying.

H T O

1 2 5

1 4 8

1 1 (Carry overs)

2 7 3

- **Ones:** 5+8=13. Write 3, carry 1 ten.
- **Tens:** 1(carried)+2+4=7. Write 7.
- **Hundreds:** 1+1=2. Write 2.

3. **Step 3:** Write the answer with the unit: **273 yams.**

EVALUATION

1. Solve: $27 + 35$.
2. Add: $156 + 126$.
3. Find the sum of 49 and 32.
4. In a class, there are 25 boys and 28 girls. How many pupils are there altogether?
5. What is 345 plus 145?
6. Musa has 88 kobo and Ali has 14 kobo. What is the total amount of money?

ASSIGNMENT

- **(Written):** Calculate: $365 + 427$. Show your working with carrying.
- **(Written):** Bola bought a school bag for ₦150 and a book for ₦60. How much did she spend altogether?
- **(Oral/Practical):** Give someone at home ₦77 in pretend notes/coins and another ₦23. Ask them to add it up and tell you the total. Check their answer.

WEEK 3: SUBTRACTION (WITHOUT BORROWING).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Subtract 2-digit numbers without borrowing.
- Subtract 3-digit numbers without borrowing.
- State that subtraction means taking away or finding the difference.

CONTENT Understanding Subtraction: Subtraction means taking away one number from another.

- It is like giving away some of your sweets and seeing how many are left.
- The sign for subtraction is the minus sign (-). We always start subtracting from the **Ones** column.

Worked Example 1: Subtracting 2-Digit Numbers (65 - 23)

1. **Step 1: Arrange.** Write the bigger number on top. Line up tens and ones.

T O
6 5
2 3

2. **Step 2: Subtract Ones.** $5 - 3 = 2$. Write 2 under the Ones column.
3. **Step 3: Subtract Tens.** $6 - 2 = 4$. Write 4 under the Tens column.
- **Answer:** 42.

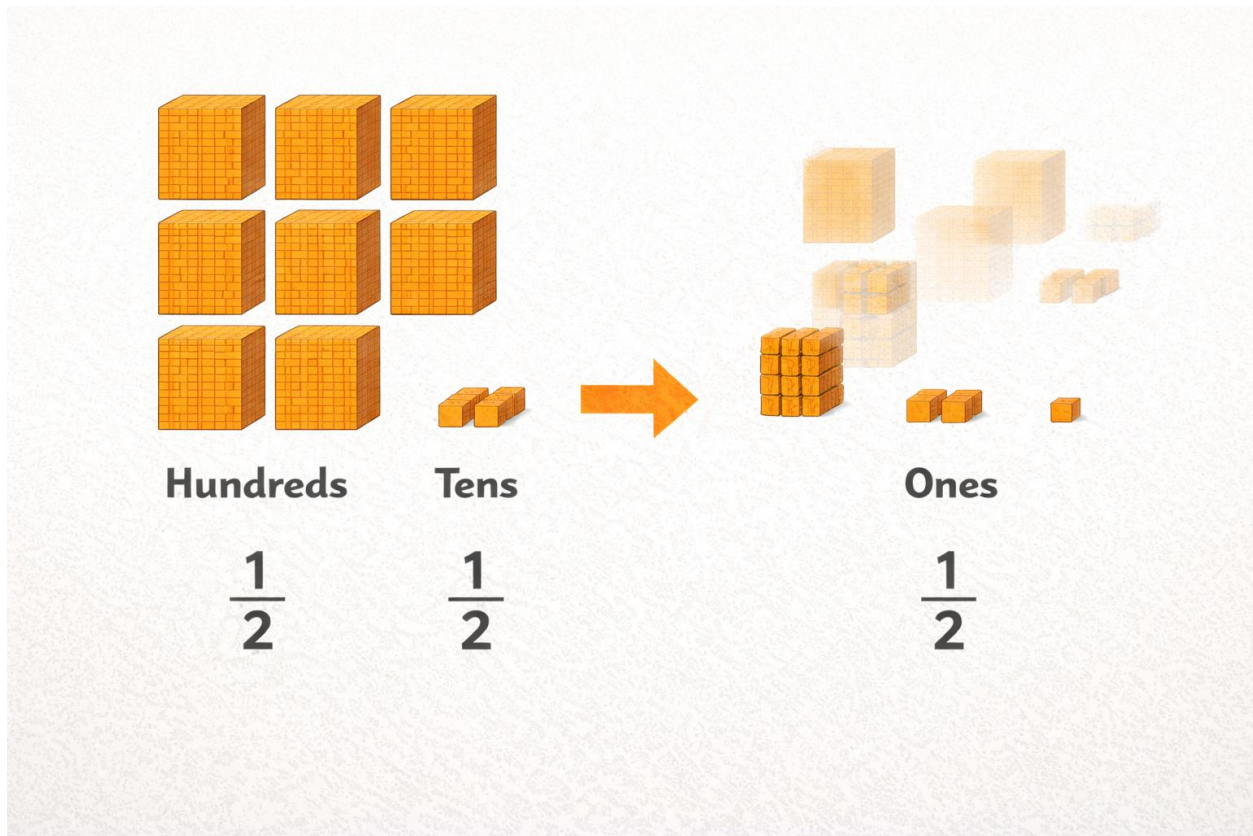
Worked Example 2: Subtracting 3-Digit Numbers (784 - 251)

1. **Step 1: Arrange.** Line up hundreds, tens, and ones.

H T O
7 8 4
2 5 1

2. **Step 2: Subtract Ones.** $4 - 1 = 3$.
3. **Step 3: Subtract Tens.** $8 - 5 = 3$.
4. **Step 4: Subtract Hundreds.** $7 - 2 = 5$.

- **Answer:** 533.



Key Words for Subtraction: take away, minus, less, difference, how many are left, remain.

EVALUATION

1. $45 - 21 = ?$
2. Subtract 10 from 50.
3. $87 - 34 = ?$
4. Solve: $695 - 243$.
5. What is 77 minus 22?
6. Remove 300 from 900.

ASSIGNMENT

- **(Written):** Subtract 42 from 96.
- **(Written):** Find the difference between 555 and 222.

- **(Practical/Oral):** You have 89 seeds. Give 50 seeds to a friend. Count how many seeds you have left. Write the subtraction sentence for it.
-



WEEK 4: SUBTRACTION (WITH BORROWING AND WORD PROBLEMS).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Subtract numbers involving "borrowing" (renaming).
- Solve word problems involving subtraction.
- Identify keywords like "difference," "remain," and "left."

CONTENT Subtraction with Borrowing: Sometimes, the top digit in a column is smaller than the bottom digit. We cannot subtract (e.g., you cannot take 8 oranges from a basket that has only 2). We need to **borrow** from the next column on the left.

- **Important Rule:** When you borrow 1 from the Tens column, it becomes 10 Ones in the Ones column.

Worked Example 1: 52 - 38

1. **Step 1: Arrange.**

T O
5 2
3 8

2. **Step 2: Look at Ones.** $2 - 8$? Impossible. We need more Ones.

3. **Step 3: Borrow from Tens.**

- Cross out the 5 in the Tens place. It becomes **4** (because we are taking 1 Ten away).
- Give the 1 Ten (which is 10 Ones) to the Ones place.
- The 2 becomes **12** ($10 + 2$).

4. **Step 4: Now subtract Ones.** $12 - 8 = 4$. Write 4 in the Ones column.

5. **Step 5: Subtract Tens.** 4 (remaining Tens) $- 3 = 1$. Write 1 in the Tens column.

- **Answer:** 14.

$$\begin{array}{r}
 52 \\
 - 4 \\
 \hline
 38
 \end{array}$$

$2 - 8$
 Cannot Subtract
 $12 - 8 = 4$

Tens	Ones
2	2
$\rightarrow$	
2	$= 12$
8	$=$
4	1

Final Answer: 14

Word Problems with Subtraction: Look for clues: "left," "remaining," "how many more," "difference," "how much is left."

Worked Example 2: There are 250 eggs in a crate. 135 eggs are broken. How many eggs are left unbroken?

- Step 1:** The word "left" tells us to subtract. Numbers: $250 - 135$.
- Step 2:** Arrange and solve. Watch for borrowing!

H T O

2 5 0 $\rightarrow$ (Borrowing steps: 0-5 impossible. Borrow from 5 Tens.)

4 10

- 1 3 5

1 1 5

- Ones:** $0 - 5$? Borrow. 5 Tens becomes 4, 0 Ones becomes 10. $10 - 5 = 5$.
- Tens:** 4 (remaining) - 3 = 1.

- **Hundreds:** $2 - 1 = 1$.
3. **Answer:** **115 eggs** are left unbroken.

EVALUATION

1. $71 - 49 = ?$
2. Subtract 28 from 60.
3. A farmer has 100 yams. He sells 45. How many yams remain?
4. Find the difference between 54 and 19.
5. $325 - 118 = ?$
6. If I have ₦50 and I spend ₦35 on *akara*, how much change do I have left?

ASSIGNMENT

- **(Written):** Solve: $92 - 67$. Show your borrowing steps.
 - **(Written):** Solve: $403 - 281$.
 - **(Oral/Practical):** There are 45 pupils in a class. If 18 are girls, how many are boys?
Explain to someone how you got your answer.
-

WEEK 5: MIDTERM TEST.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Attempt questions covering Weeks 1--4 independently.

CONTENT Assessment Scope: The test will cover all you have learned about Addition and Subtraction this term.

- Addition of 2-digit and 3-digit numbers **without** carrying.
- Addition of 2-digit and 3-digit numbers **with** carrying.
- Subtraction of 2-digit and 3-digit numbers **without** borrowing.
- Subtraction of 2-digit and 3-digit numbers **with** borrowing.
- Solving simple word problems using addition and subtraction.

Test Tips:

- Read each problem twice.
- For word problems, underline the clue words (like "total" or "left").
- Always check if you need to carry or borrow.
- Review your answers before submitting.

EVALUATION

1. The teacher administers the written test.
2. Post-test, the teacher guides a class correction session to address common errors.

ASSIGNMENT

- **(Written):** From your test, choose one word problem you solved correctly. Write it out again and explain your steps to a family member.
- **(Written):** Correct any two calculation errors from your test paper. Write the correct working neatly.
- **(Oral):** Teach a younger sibling how to do a simple addition like $24+13$ without carrying.

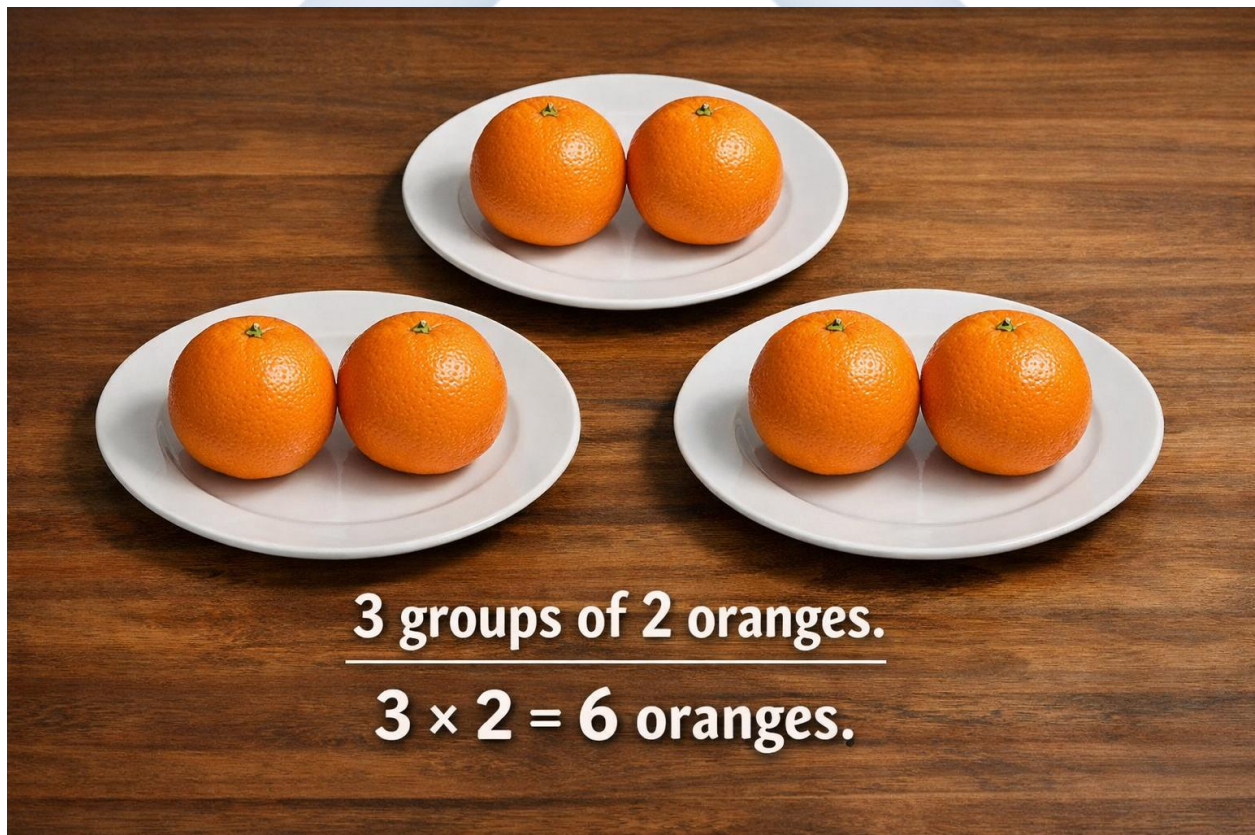
WEEK 6: MULTIPLICATION (REPEATED ADDITION).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define multiplication as repeated addition.
- Use the symbol ' $\times$ ' for multiplication.
- Recite multiplication tables for 2, 3, 4, and 5.

CONTENT Multiplication is Repeated Addition: Multiplication is a faster way to add the same number many times.

- **Example:** $2 + 2 + 2$ is adding 2 three times.
- We say: "**3 groups of 2**" or "**3 times 2**".
- We write it as: $3 \times 2 = 6$. The ' $\times$ ' is the multiplication sign.
- It means "groups of."



Multiplication Tables (Facts): Learning your "times tables" helps you multiply quickly. Let's start with tables for 2, 3, 4, and 5.

- **2 Times Table:** (Think of pairs: shoes, hands, eyes)

- $1 \times 2 = 2$ (One group of 2)
- $2 \times 2 = 4$ (Two groups of 2: $2+2$)
- $3 \times 2 = 6$ (Three groups of 2: $2+2+2$)
- $4 \times 2 = 8$
- $5 \times 2 = 10$

- **3 Times Table:** (Think of triplets or triangle sides)

- $1 \times 3 = 3$
- $2 \times 3 = 6$ ($3+3$)
- $3 \times 3 = 9$ ($3+3+3$)
- $4 \times 3 = 12$
- $5 \times 3 = 15$

Worked Example (Word Problem): If one tricycle (keke) has 3 wheels, how many wheels do 4 tricycles have?

- **Addition way:** $3 + 3 + 3 + 3 = 12$ wheels.
- **Multiplication way:** 4 groups of 3 wheels = $4 \times 3 = 12$ wheels.

EVALUATION

1. Write $5 + 5 + 5$ as a multiplication sentence.
2. What is 4×2 ?
3. What is 5×3 ?
4. $2 \times 6 = ?$ (Think: two groups of six)

5. Complete the pattern from the 3 times table: 3, 6, 9, ___, ___.
6. How many legs do 2 dogs have? (A dog has 4 legs). Write it as a multiplication.

ASSIGNMENT

- **(Written):** Write out the 4 times table from 1×4 to 5×4 in your notebook.
- **(Written):** Change " $3 + 3 + 3 + 3$ " to a multiplication sum. What is the answer?
- **(Oral/Practical):** Recite the 5 times table (1×5 to 5×5) to someone at home. Use your fingers to count the groups.



WEEK 7: MIDTERM BREAK.



WEEK 8: MULTIPLICATION (2-DIGIT BY 1-DIGIT).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Multiply a 2-digit number by a 1-digit number without carrying.
- Multiply a 2-digit number by a 1-digit number with carrying.
- Solve simple word problems on multiplication.

CONTENT Method: Vertical Multiplication: We can multiply bigger numbers by writing them in columns, just like addition. We multiply the **Ones** place first, then the **Tens** place.

Worked Example 1: No Carrying (23×2)

1. **Step 1: Arrange.** Write the 2-digit number on top, the 1-digit number below, aligned to the ones.

$$\begin{array}{r} \text{T O} \\ 23 \\ \times 2 \\ \hline \end{array}$$

2. **Step 2: Multiply Ones.** $3 \text{ (ones)} \times 2 = 6$. Write 6 in the Ones place of the answer.
 3. **Step 3: Multiply Tens.** $2 \text{ (tens)} \times 2 = 4$. Write 4 in the Tens place of the answer.
- **Answer:** 46.

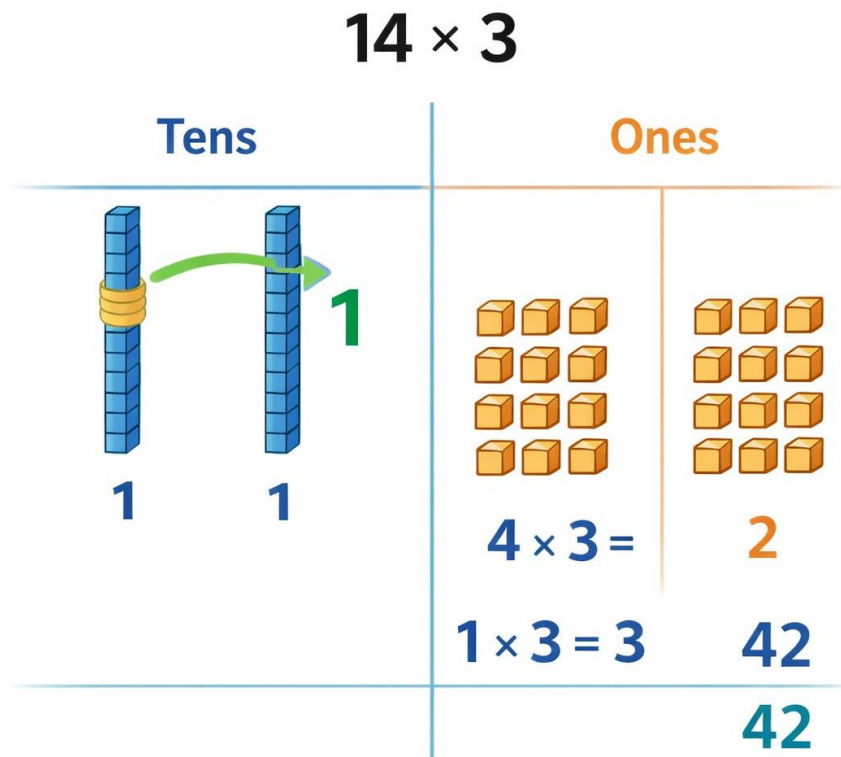
Worked Example 2: With Carrying (14×3)

1. **Step 1: Arrange.**

$$\begin{array}{r} \text{T O} \\ 14 \\ \times 3 \\ \hline \end{array}$$

2. **Step 2: Multiply Ones.** $4 \times 3 = 12$.
 - 12 means 1 Ten and 2 Ones.
 - Write **2** in the Ones place.
 - **Carry** the **1** (Ten) to the top of the Tens column (write a small '1').
3. **Step 3: Multiply Tens.** $1 \text{ (Ten)} \times 3 = 3$.

- Now **add** the carried 1: $3 + 1 = 4$.
- Write 4 in the Tens place.
- **Answer:** 42.



Word Problems with Multiplication: Key words: "times," "product," "each," "groups of," "altogether" (when same amount is repeated).

Example: One pack of biscuits contains 12 biscuits. How many biscuits are in 4 such packs?

- This is 4 groups of 12. 12×4 .
- Calculate: $12 \times 4 = 48$ biscuits.

EVALUATION

1. $21 \times 3 = ?$
2. $12 \times 4 = ?$
3. Multiply 15 by 2.

4. Solve: 33×3 .
5. Find the product of 11 and 5.
6. One exercise book costs ₦20. How much will 3 of the same books cost?

ASSIGNMENT

- **(Written):** Multiply 24 by 2.
 - **(Written):** Multiply 16 by 3. Show any carrying.
 - **(Oral/Practical):** A wheelbarrow has 1 wheel. A bicycle has 2 wheels. How many wheels do 5 bicycles have? Write the multiplication sentence and solve.
-



WEEK 9: DIVISION (SHARING AND RELATION TO MULTIPLICATION).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define division as sharing things equally.
- Identify the division sign ($\div$).
- Relate division to multiplication (Inverse relationship).

CONTENT What is Division?: Division means sharing or splitting a number of things into equal groups or among equal people.

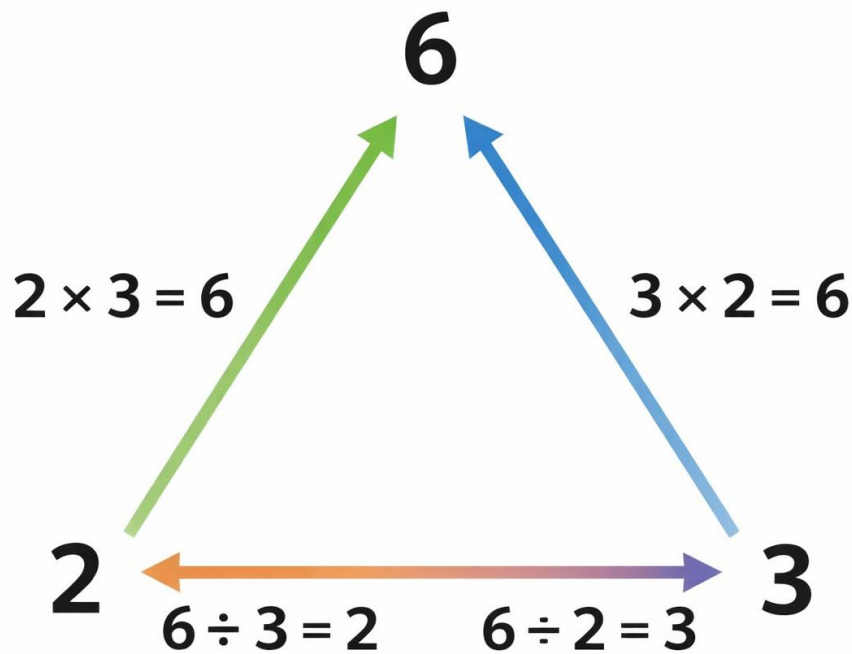
- It is the opposite of putting groups together (multiplication). The sign for division is $\div$.
- It means "shared by" or "divided into groups of."

Example: Share 6 sweets equally between 2 friends.

- Give one sweet to each friend, again and again.
- Each friend gets 3 sweets.
- We write: $6 \div 2 = 3$.

Multiplication and Division are Best Friends (Inverse Operations): If you know your multiplication facts, division becomes easy! They undo each other.

- If $2 \times 3 = 6$, then $6 \div 2 = 3$ and $6 \div 3 = 2$.
- If $4 \times 5 = 20$, then $20 \div 4 = 5$ and $20 \div 5 = 4$.



Worked Example (Using Multiplication to Divide): Share 15 oranges equally among 3 baskets.
How many oranges in each basket?

- Think: What number, when multiplied by 3, gives 15? From our times tables: $3 \times 5 = 15$.
- So, $15 \div 3 = 5$.
- **Answer:** Each basket gets **5 oranges**.

EVALUATION

1. What does the division sign ($\div$) mean?
2. Share 8 oranges equally between 2 people. How many does each get?
3. $10 \div 2 = ?$
4. If $3 \times 4 = 12$, what is $12 \div 3$?
5. Divide 20 by 5.
6. There are 12 pencils to be shared equally among 4 pupils. How many pencils does each pupil get?

ASSIGNMENT

- **(Written):** Solve: $16 \div 2$.
 - **(Written):** Solve: $15 \div 5$.
 - **(Practical/Oral):** Use 18 seeds or pebbles. Share them equally into 3 groups. How many are in each group? Write the division sentence for what you did.
-



WEEK 10: DIVISION (WITH REMAINDERS).

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify simple division problems where items remain.
- Solve division problems with remainders.
- Write the answer as "Quotient remainder Remainder".

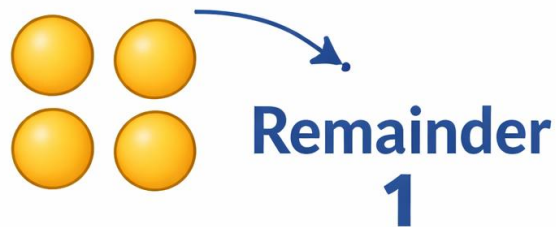
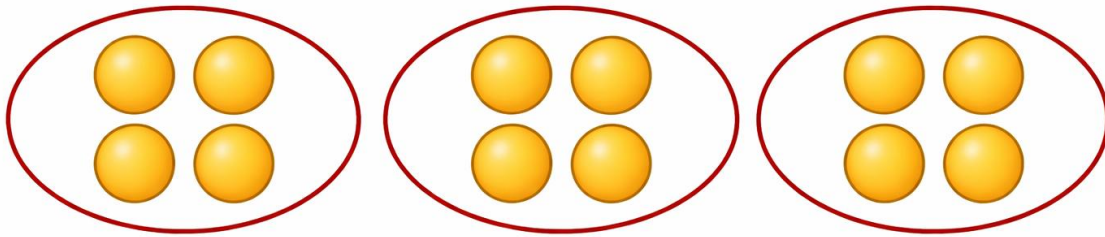
CONTENT What is a Remainder?: Sometimes, when we try to share things equally, there are some items left over that cannot form a complete group. These left-over items are called the **remainder**.

Worked Example 1: Sharing with a Remainder: Share 7 balls between 2 boys equally.

1. Give 1 ball to Boy A, 1 ball to Boy B. (2 balls used)
 2. Give another 1 to Boy A, another 1 to Boy B. (4 balls used)
 3. Give another 1 to Boy A, another 1 to Boy B. (6 balls used)
 4. There is **1 ball left**. It cannot be shared equally because we need 2 more to give one each.
- **Answer:** Each boy gets **3 balls**, and there is **1 ball left over**.
 - We write: $7 \div 2 = 3 \text{ remainder } 1$.

Worked Example 2: Using Multiplication to Find Remainders ($13 \div 4$)

1. **Step 1:** Think of the 4 times table: 4, 8, 12, 16...
 2. **Step 2:** Find the biggest number that is **less than or equal to 13**. That is **12** (because $4 \times 3 = 12$).
 3. **Step 3:** The **3** is how many full groups we can make. This is the **quotient**.
 4. **Step 4:** Find what's left: $13 - 12 = 1$. This '1' is the **remainder**.
- **Answer:** **3 remainder 1**.



EVALUATION

1. Divide 5 by 2. What is the answer and remainder?
2. Share 10 sweets equally among 3 children. How many sweets are left over?
3. $17 \div 5 = ?$ (Find the quotient and remainder).
4. Divide 9 by 4.
5. If I share 13 mangoes equally between 2 friends, how many mangoes will be left (remainder)?
6. $21 \div 5 = ?$

ASSIGNMENT

- **(Written):** Solve: $11 \div 2$. Write your answer with the remainder.
- **(Written):** Solve: $14 \div 3$.

- **(Drawing/Practical):** Draw 7 circles. Try to group them in 2s by circling them. How many groups of 2 did you make? How many circles are left over? Write the division sentence.
-



WEEKS 11-13: REVISION AND EXAMINATION.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Recall and solve questions from all topics covered in the term.

CONTENT Comprehensive Revision: We will go over all the major topics from this term:

- **Addition:** With and without carrying. Solving word problems.
- **Subtraction:** With and without borrowing. Solving word problems.
- **Multiplication:** As repeated addition. Multiplying 2-digit by 1-digit (with/without carry). Times tables (2,3,4,5).
- **Division:** As equal sharing. Relationship with multiplication. Division with remainders.

Examination Preparation Tips:

- Practice setting out your work neatly in columns.
- For word problems, identify the correct operation (+, -, $\times$, $\div$) first.
- Review your times tables.
- Get a good rest before the exam day.

EVALUATION

- **Weeks 11-12:** Guided revision sessions with practice questions.
- **Week 13:** End-of-Term Examination covering all Second Term topics.

ASSIGNMENT

- **(Written):** Create one practice question each for addition, subtraction, and multiplication. Solve them.
- **(Oral):** Ask a family member to test you on 10 random multiplication facts from the 2, 3, 4, and 5 times tables.
- **(Review):** Look through your notebook and identify one topic you want to practice more. Do 3 extra questions on that topic.

PRIMARY 3 MATHEMATICS SCHEME OF WORK

THIRD TERM

Week	Topic	Content Breakdown
1	Open Sentences	Completing missing numbers in addition and subtraction sentences (e.g., $12 + \underline{\quad} = 20$).
2	Money	Identifying Nigerian coins and notes; counting, adding and subtracting money.
3	Length I	Measuring objects using non-standard units (hand span, sticks).
4	Length II	Measuring using standard units (metres, centimetres); comparing lengths.
5	Midterm Test	—
6	Time	Reading time to the hour, half-hour, and quarter-hour on clocks.
7	Midterm Break	—
8	Weight	Measuring objects using standard units (grams, kilograms); comparing weights.
9	Capacity	Measuring liquids using standard units (litres, millilitres).
10	Symmetry	Identifying and drawing symmetrical shapes; folding to find lines of symmetry.
11	Pictograms	Collecting data, representing data using pictograms, interpreting simple pictographs.
12–13	Revision / Examination / Closing	—

PRIMARY 3 MATHEMATICS LESSON NOTES

THIRD TERM

WEEK 1: OPEN SENTENCES.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define an open sentence.
- Find missing numbers in addition sentences (e.g., $5 + \square = 12$).
- Find missing numbers in subtraction sentences (e.g., $15 - \square = 10$).

CONTENT What is an Open Sentence?: An open sentence is like a puzzle in mathematics.

- It is a number sentence that has a missing part.
- The missing part is like a hidden treasure we need to find.
- It is often shown by a box ($\square$), a line ($\underline{\quad}$), or a letter like a or x .
- Our job is to find the number that makes the sentence true and complete.

Finding Missing Numbers in Addition Sentences: When the missing number is one of the numbers being added, we use subtraction to find it. Remember, addition and subtraction are opposites.

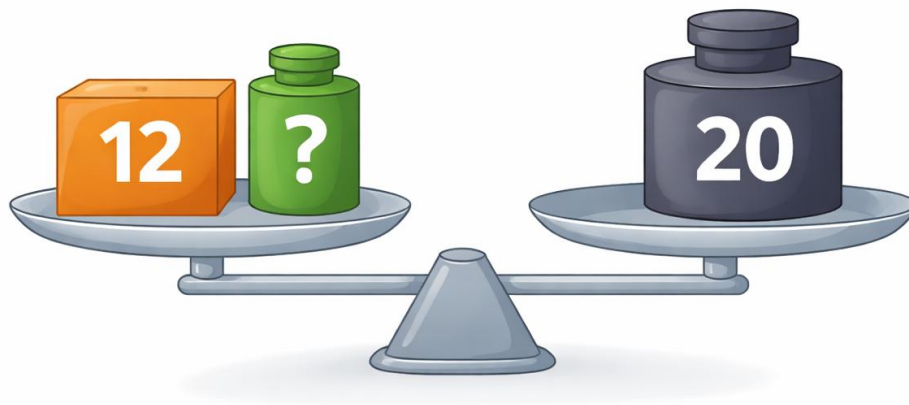
- **Rule:** To find a missing addend, **subtract** the known addend from the total (sum).

Example 1: $12 + \square = 20$

- Think: "12 plus what number equals 20?"
- Method: $20 - 12 = 8$.
- **Answer:** $\square = 8$ (We can check: $12 + 8 = 20$ ✓).

Example 2: $\square + 5 = 15$

- Think: "What number plus 5 equals 15?"
- Method: $15 - 5 = 10$.
- **Answer:** $\square = 10$ (Check: $10 + 5 = 15$ ✓).



**To find “?”,
take away 12 from 20.**

Finding Missing Numbers in Subtraction Sentences: There are two types of missing numbers in subtraction: the number being subtracted (taken away) and the number we start with.

Case 1: Missing Middle Number (e.g., $20 - \square = 15$)

- **Rule:** Subtract the answer from the first number.
- **Method:** $20 - 15 = 5$.
- **Answer:** $\square = 5$ (Check: $20 - 5 = 15$ ✓).

Case 2: Missing First Number (e.g., $\square - 4 = 10$)

- **Rule:** Add the two smaller numbers together.
- **Method:** $10 + 4 = 14$.
- **Answer:** $\square = 14$ (Check: $14 - 4 = 10$ ✓).

Sentence Type	How to Solve	Example
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Sentence Type	How to Solve	Example
Big - ? = Small	Big - Small = ?	$20 - ? = 15 \rightarrow 20 - 15 = 5$
? - Small = Big	Small + Big = ?	$? - 4 = 10 \rightarrow 4 + 10 = 14$

EVALUATION

- $7 + \square = 15$
- $\square + 10 = 30$
- $18 - \square = 12$
- $\square - 5 = 20$
- $9 + 9 = \square$
- $25 - \square = 20$

ASSIGNMENT

- (Written):** Find the missing number: $35 + \square = 50$.
- (Written):** Find the missing number: $\square - 10 = 40$.
- (Practical/Oral):** Give someone at home this puzzle: "I have some oranges. If I give you 7, I will have 13 left. How many oranges did I have at first?" Help them write and solve the open sentence ($\square - 7 = 13$).

WEEK 2: MONEY.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify Nigerian coins (kobo) and bank notes (Naira).
- Count sums of money correctly.
- Add sums of money together.
- Subtract sums of money to find change.

CONTENT Identification of Nigerian Currency: We use money called Naira (₦) and Kobo (K). 100 Kobo = 1 Naira.

- **Common Coins:** 50K, ₦1, ₦2.
- **Common Bank Notes:** ₦5, ₦10, ₦20 (Green), ₦50 (Blue), ₦100 (Red/Brown), ₦200, ₦500, ₦1000.



Counting Money: To count money, sort notes and coins by their value. Start with the biggest value (e.g., ₦500 notes) and move to the smallest (e.g., 50K coins).

- **Example:** ₦100 + ₦50 + ₦20 + ₦5 = ₦175.

Adding Money: We add money just like ordinary numbers. Remember to include the naira sign (₦) in your answer.

- **Example:** If a textbook costs ₦250 and a notebook costs ₦150, the total cost is $₦250 + ₦150 = ₦400$.

Subtracting Money (Finding Change): When you buy something and pay with more money than the price, the seller gives you back the difference. This is called **change**.

- **Example:** You buy a *suya* for ₦120. You give the seller a ₦200 note.
- Your change is $₦200 - ₦120 = ₦80$.



To find “?”, take away 12 from 20.

EVALUATION

1. What is the colour of the ₦50 note?
2. Add: $₦100 + ₦50 + ₦20$.
3. How many ₦10 notes make ₦50?
4. Subtract: $₦90 - ₦40$.
5. You buy a drink for ₦30. If you pay with a ₦50 note, how much change should you get?
6. List three Nigerian bank notes and their colours.

ASSIGNMENT

- **(Written):** Find the total: ₦200 + ₦100 + ₦50.
 - **(Written):** Chioma has ₦500. She spends ₦200 on cloth. How much money does she have remaining?
 - **(Practical/Drawing):** Draw two items you can buy at home (e.g., soap, bread). Write a price under each (less than ₦100). Add the prices. If you pay with ₦100, how much change will you get? Draw the change as coins or notes.
-



WEEK 3: LENGTH I (NON-STANDARD UNITS).

OBJECTIVES By the end of the lesson, pupils should be able to:

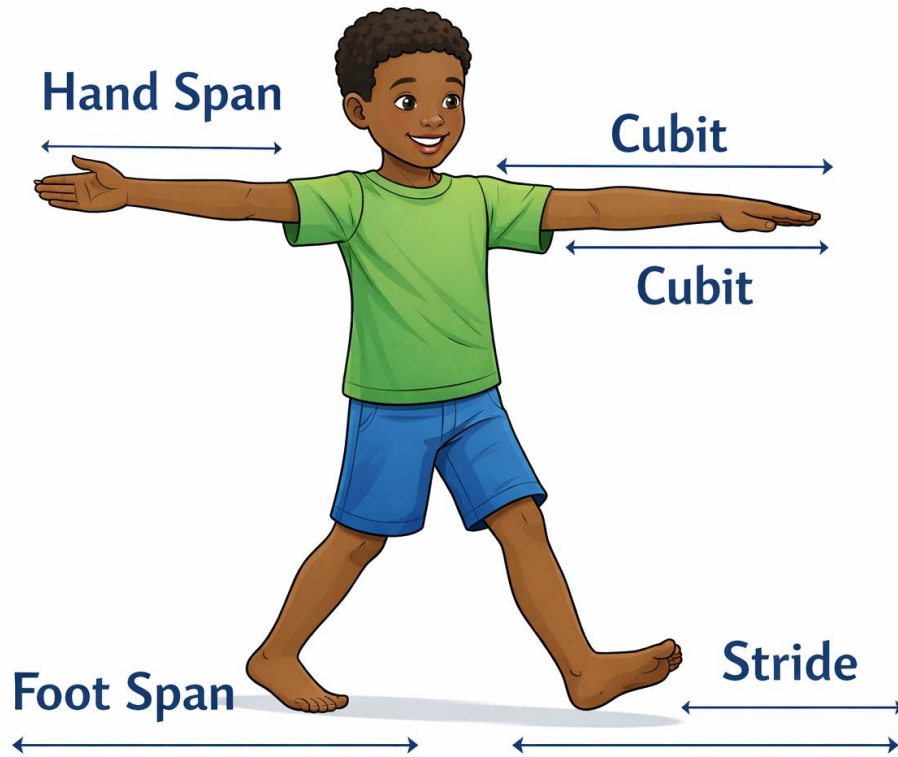
- Define length as the distance from one point to another.
- List three non-standard units of measurement.
- Measure classroom objects using hand spans and strides.

CONTENT What is Length?: Length is how long, tall, or far something is.

- It is the distance from one end of an object to the other end.
- We measure length to compare sizes (e.g., which stick is longer?).

Non-Standard Units: Long ago, people used parts of their own bodies to measure things because they always had them with them! These are called **non-standard units** because they are not the same for everyone.

- **Hand Span:** The distance from the tip of your thumb to the tip of your little finger when your hand is stretched out.
- **Foot Span:** The length of your foot from heel to toe.
- **Cubit:** The distance from your elbow to the tip of your middle finger.
- **Stride (Pace):** The length of one normal step when you walk.
- **Arm Span:** The distance from the tip of your left middle finger to the tip of your right middle finger when your arms are stretched wide.



Measuring with Non-Standard Units: Let's measure using our hand spans.

- **Activity:** How many of your hand spans is your desk?
 1. Place your hand at one end of the desk.
 2. Mark where your little finger reaches.
 3. Move your hand to that mark and repeat.
 4. Count how many hand spans until you reach the other end.
- **Important:** Your teacher's hand span is bigger than yours. If you both measure the same table, you will get a **larger number** of hand spans because your hand is smaller.

Why "Non-Standard"?: These units are not fixed. My hand span is different from yours. So, we need a fixed, standard way to make sure everyone gets the same measurement.

EVALUATION

1. What do we mean by "length"?
2. Mention two body parts that can be used to measure length.

3. Use your hand span to measure the width of your math textbook. How many spans was it?
4. Measure the length of the classroom door using your foot spans.
5. Why is a "hand span" called a non-standard unit?

ASSIGNMENT

- **(Practical):** At home, measure the length of your bed using your hand span. Record the number.
 - **(Practical):** Measure the length of your sitting room or veranda using your strides (steps). Write down the number of strides.
 - **(Drawing):** Draw an outline of your hand and label it "My Hand Span for Measuring".
-



WEEK 4: LENGTH II (STANDARD UNITS).

OBJECTIVES By the end of the lesson, pupils should be able to:

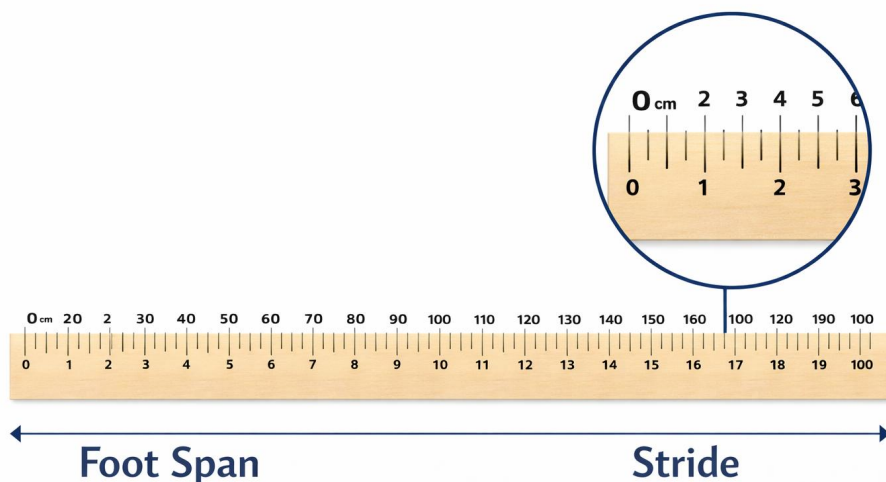
- Identify the standard units for measuring length (Metre and Centimetre).
- Use a ruler to measure objects in centimetres (cm).
- State the relationship between Metre and Centimetre ($100\text{cm} = 1\text{m}$).

CONTENT Standard Units: To make sure everyone, everywhere, gets the same measurement, we use **standard units** and special tools.

- **Centimetre (cm):** A small unit. Used to measure short lengths like a pencil, a finger, or a book. We use a **ruler** to measure in cm.
- **Metre (m):** A bigger unit. Used to measure longer lengths like a table, a room, or a piece of cloth. We use a **metre rule** or a **tape measure**.

Relationship Between Metre and Centimetre:

- 1 metre (m) = 100 centimetres (cm).
- Think of it like this: One big metre stick is the same length as 100 small centimetre sticks placed end-to-end.



$$100\text{ cm} = 1\text{ m}$$

How to Use a Ruler

1. Place the object you are measuring so one end is at the **0 mark** on the ruler.
 2. Do not start from the very end of the ruler.
 3. Look straight down at the ruler to see where the other end of the object reaches.
 4. Read the number on the ruler at that point. That number is the length in **centimetres (cm)**.
- **Example:** A pencil's tip is at 0 cm. Its eraser end is at 15 cm. The pencil is **15 cm** long.

Comparing Lengths

- Which is longer: 50 cm or 1 m?
- Since 1 m = 100 cm, 100 cm is more than 50 cm. So, **1 m is longer**.
- 200 cm = ____ m?
- Since 100 cm = 1 m, then 200 cm = 2 m.

EVALUATION

1. What tool do we use to measure the length of a line in our book?
2. How many centimetres make 1 metre?
3. Which unit is bigger: cm or m?
4. Use your ruler to measure the length of your pen or pencil in cm.
5. Would you measure the height of a door in cm or m?
6. Complete: 300 cm = ____ m.

ASSIGNMENT

- **(Written/Practical):** Use a ruler to measure the length of three different items in your house (e.g., spoon, key, toothbrush). Write down each item and its length in cm.
- **(Written):** Which is longer: a 30 cm ruler or a 1 m stick? Explain your answer.
- **(Oral):** Tell a family member what 'cm' and 'm' stand for, and name one thing you would measure with each unit.

WEEK 5: MIDTERM TEST.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Answer questions covering Weeks 1--4 independently.

CONTENT Assessment Focus: The test will check your understanding of:

- **Open Sentences:** Finding missing numbers in addition and subtraction.
- **Money:** Identifying notes/coins, adding and subtracting amounts of money, and calculating change.
- **Length:** Understanding non-standard and standard units (cm, m), and how to measure with a ruler.

Test Strategy:

- Read each question slowly.
- For money questions, remember to write the naira sign (₦).
- For measurement, think: is this a long or short object? Should I use cm or m?
- Check your work for simple mistakes.

EVALUATION

1. Teacher administers the written test.
2. Followed by a class discussion to review correct answers and clarify misunderstandings.

ASSIGNMENT

- **(Written):** From your test, pick one question you found interesting. Write it down and explain why it was interesting.
- **(Written):** Find one mistake in your test (if any) and write out the correct working and answer.
- **(Practical):** Teach someone at home how to measure the length of a table using a ruler or tape measure. Explain the unit you are using (cm).

WEEK 6: TIME.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Identify the long hand (minute) and short hand (hour) on a clock face.
- Read and tell the time to the Hour (e.g., 3 o'clock).
- Read and tell the time to the Half-hour (e.g., Half past 3).
- Read and tell the time to the Quarter-hour (e.g., Quarter past/to).

CONTENT Parts of an Analogue Clock: The face of a clock has two main hands:

- **Short Hand (Hour Hand):** This hand is shorter and fatter. It tells us the **hour**. It moves slowly from one number to the next.
- **Long Hand (Minute Hand):** This hand is longer and thinner. It tells us the **minutes**. It makes a full circle every 60 minutes.



Reading the Time

- **O'Clock:** When the long minute hand points straight up to the **12**, it is exactly an hour.
- Example: Short hand on 3, long hand on 12 = **3 o'clock** (3:00).

- **Half Past:** When the long minute hand points straight down to the **6**, it is halfway around the clock. The short hand will be halfway between two numbers.
- Example: Short hand between 4 and 5, long hand on 6 = **Half past 4** (4:30).
- **Quarter Past:** When the long minute hand points to the **3**, it is a quarter (15 minutes) past the hour.
- Example: Short hand just past 2, long hand on 3 = **Quarter past 2** (2:15).
- **Quarter To:** When the long minute hand points to the **9**, it is a quarter (15 minutes) to the next hour. We look at the **next** hour number.
- Example: Short hand almost at 5, long hand on 9 = **Quarter to 5** (4:45). This means 15 minutes before 5 o'clock.



3:00 (O'Clock)



4:30 (Half Past)



2:15 (Quarter Past)



4:45 (Quarter To)

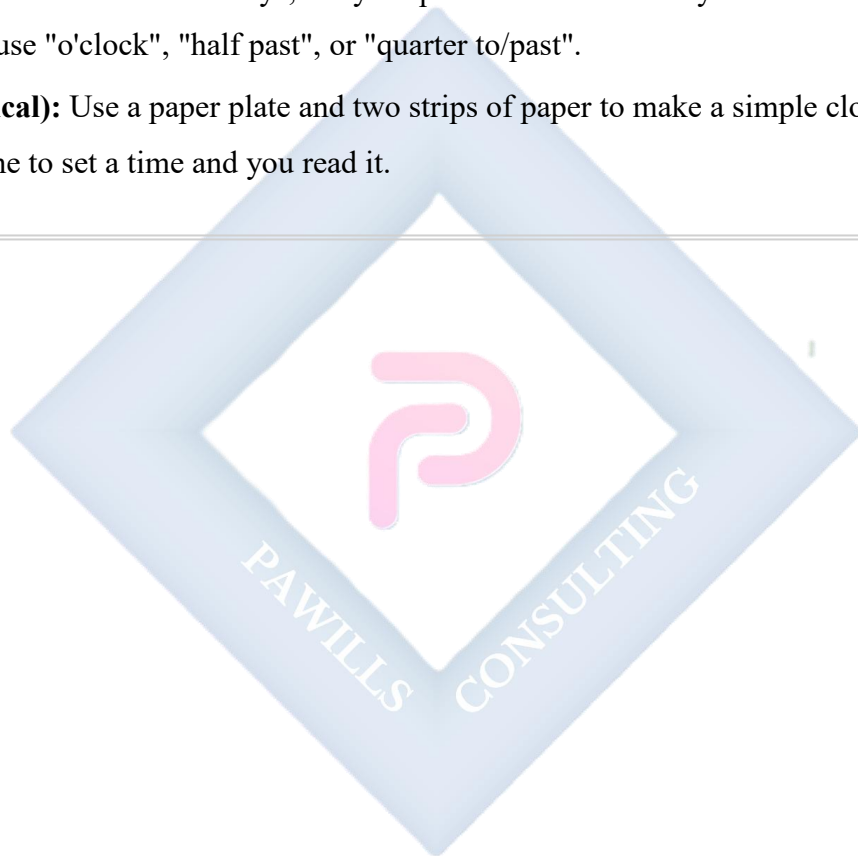
EVALUATION

1. On a clock, which hand tells us the hour?
2. What time is it when the long hand is on 12 and the short hand is on 9?
3. Draw a clock showing 2 o'clock.

4. Where does the long hand point for "Half Past"?
5. Read this time: Short hand just past 10, long hand on 3.
6. If the time is "Quarter to 7", where is the long hand?

ASSIGNMENT

- **(Drawing):** Draw clock faces to show these times: a) 5 o'clock. b) Half past 8. c) Quarter past 12.
 - **(Oral):** For the next two days, tell your parent the time when you sit down to eat dinner. Try to use "o'clock", "half past", or "quarter to/past".
 - **(Practical):** Use a paper plate and two strips of paper to make a simple clock. Ask someone to set a time and you read it.
-



WEEK 7: MIDTERM BREAK.



WEEK 8: WEIGHT.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define weight as how heavy or light an object is.
- Identify standard units of weight (Grams and Kilograms).
- Compare the weight of different objects using a balance scale.

CONTENT What is Weight?: Weight tells us how heavy something is.

- It is the force with which the earth pulls an object down. We feel weight when we lift things.
- A feather is **light**.
- A sack of rice is **heavy**.

Standard Units of Weight: We measure weight in standard units so everyone agrees.

- **Gram (g):** Used for very light things. Examples: a sheet of paper, a pencil, a key.
- **Kilogram (kg):** Used for heavier things. Examples: a bag of sugar, a bunch of plantains, a child.
- **Relationship: 1 Kilogram (kg) = 1000 Grams (g).**

Comparing Weights with a Balance Scale: A balance scale has two pans. We put objects on the pans to see which is heavier.

- If Pan A goes **DOWN** and Pan B goes **UP**, the object in Pan A is **heavier**.
- If both pans stay **LEVEL**, the objects have the **same weight**.

BALANCE SCALE SCENARIOS



Left is heavier.



Right is heavier.



Equal weight.

Real-Life Weight

- A 5-year-old child might weigh about 20 kg.
- A sachet of pure water weighs about 500 g (which is half a kilogram).

EVALUATION

1. What unit would you use to measure a bag of beans? (g or kg?)
2. What unit would you use to measure a single *maltina* bottle cap?
3. Which is heavier: a stone or a leaf?
4. How many grams make 1 kilogram?
5. What is the name of the tool we use to compare weights?
6. A bag of salt is 1 kg. A bag of cotton wool is also 1 kg. Which is heavier? (Trick! They weigh the same, but the salt is much smaller).

ASSIGNMENT

- **(Written):** List three items in your kitchen that are heavy (measured in kg).
- **(Written):** List three items in your school bag that are light (measured in g).

- **(Practical/Oral):** Find two fruits of different sizes (e.g., an orange and a lemon). Hold one in each hand. Which feels heavier? Tell someone your prediction. Then, if you have a kitchen scale, check to see if you were right!
-



WEEK 9: CAPACITY.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define capacity as the amount of liquid a container can hold.
- Identify standard units of capacity (Litres and Millilitres).
- Compare the capacity of different containers.

CONTENT

What is Capacity?: Capacity is the amount of liquid (like water, juice, oil) or sometimes dry goods (like rice) that a container can hold.

- It tells us how much space is inside.
- A teaspoon has a small capacity.
- A bucket has a large capacity.

Standard Units of Capacity: We measure capacity in:

- **Millilitre (ml):** Used for small amounts. Examples: a teaspoon of medicine (5 ml), a small water bottle (500 ml), a small paint bottle.
- **Litre (l):** Used for large amounts. Examples: a big bottle of soft drink (1.5 l or 2 l), a kettle, a jerrycan of water.
- **Relationship: 1 Litre (l) = 1000 Millilitres (ml).**



Comparing Capacities

- Which holds more: a cup or a bucket? **A bucket.**
- Which holds less: a jug or a tanker? **A jug.**
- We often use measuring cups, jugs with markings, or cylinders to measure liquids accurately.

Capacity in Daily Life

- When your mother buys vegetable oil, it is often in 1-litre, 2-litre, or 4-litre bottles.
- A sachet of pure water contains 500 ml (half a litre).

EVALUATION

1. What does capacity measure?
2. Which container has a larger capacity: a bucket or a glass?
3. What unit (ml or l) is used for a dose of cough syrup?
4. How many millilitres are in 1 litre?
5. Name two liquids you can measure in litres.
6. If a jug holds 5 cups of water and 1 cup holds 250 ml, what is the approximate capacity of the jug in ml? (5×250 ml).

ASSIGNMENT

- **(Practical):** Find an empty bottle or packet at home (e.g., milk, juice, detergent). Look for the capacity written on the label. Write down the name of the product and its capacity (e.g., "Peak Milk - 400 ml").
 - **(Drawing):** Draw a cup and a bucket. Circle the one that has the larger capacity.
 - **(Oral):** Explain to a sibling the difference between "weight" (how heavy) and "capacity" (how much it can hold). Give an example for each.
-

WEEK 10: SYMMETRY.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define symmetry.
- Identify symmetrical shapes (shapes that can be divided into two equal halves).
- Draw the line of symmetry on simple shapes.

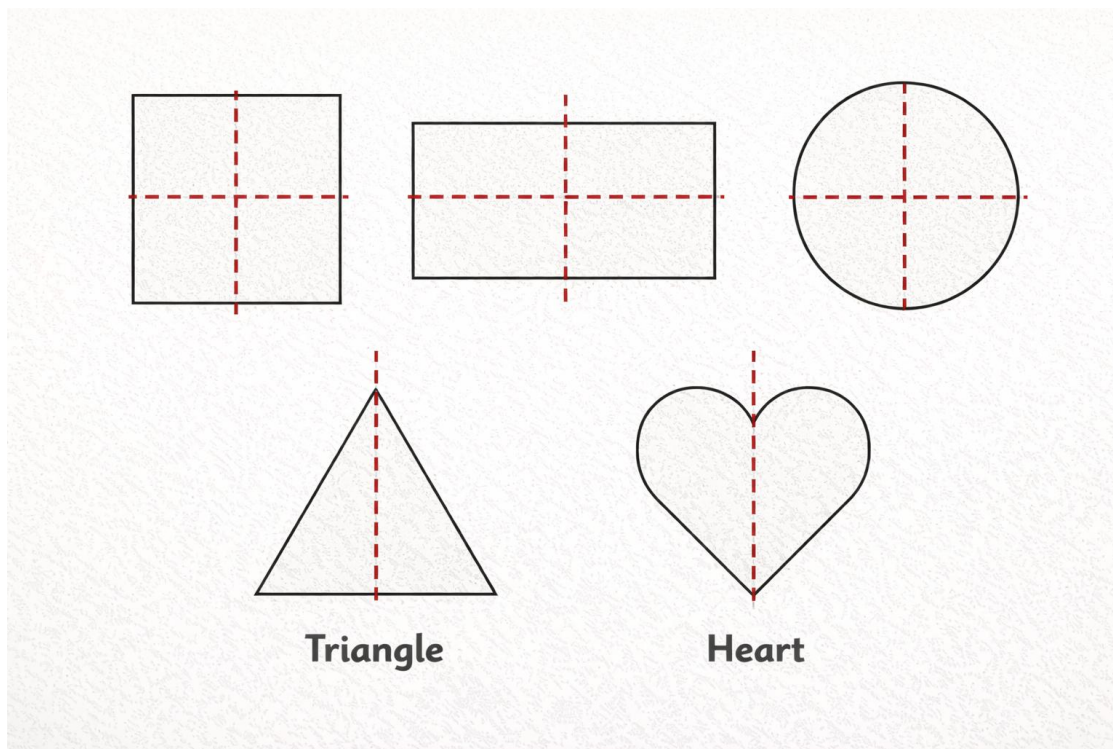
CONTENT

What is Symmetry?: Symmetry is when one half of an object is a mirror image of the other half.

- If you can fold a shape and both sides match perfectly, then it is symmetrical.
- The fold line is called the **Line of Symmetry**.

Finding Lines of Symmetry

- **Human Face:** Imagine a line down the middle of your face from forehead to chin. The two sides are roughly symmetrical.
- **Butterfly:** A line down the middle of its body divides its wings into two identical halves.
- **Shapes:**
 - **Square:** Has 4 lines of symmetry (vertical, horizontal, two diagonals).
 - **Rectangle:** Has 2 lines of symmetry (vertical and horizontal through the centre).
 - **Circle:** Has infinite lines of symmetry (any line through the centre).
 - **Equilateral Triangle:** Has 3 lines of symmetry.



The Folding Test: A simple way to check for symmetry is to imagine folding the shape along a line. If the two parts sit exactly on top of each other, that line is a line of symmetry.

EVALUATION

1. What do we call the line that divides a symmetrical shape into two matching halves?
2. How many lines of symmetry does a square have?
3. Draw a rectangle and draw all its lines of symmetry.
4. Is a regular star shape usually symmetrical?
5. Look at the letter 'A'. Is it symmetrical? Draw its line of symmetry.
6. Which shape has more lines of symmetry: a circle or a rectangle?

ASSIGNMENT

- **(Drawing):** Draw a simple butterfly. Draw its line of symmetry down the middle.
- **(Drawing):** Find and draw three symmetrical objects you can see in your classroom or at home (e.g., a window, a book, a plate).

- **(Practical):** Cut out a shape from a piece of paper (like a heart or a simple tree). Fold it to find its line(s) of symmetry. Paste it in your notebook and draw the line.
-



WEEK 11: PICTOGRAMS.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Define a pictogram as representing data using pictures.
- Collect simple data (e.g., favourite fruit).
- Represent data on a chart using pictures.
- Interpret a simple pictogram.

CONTENT

What is a Pictogram?: A pictogram (or picture graph) is a special chart that uses pictures or symbols to show information or data.

- It makes numbers easy to see and understand at a glance.

Parts of a Pictogram:

- **Title:** Tells us what the chart is about (e.g., "Favourite Fruits in Primary 3").
- **Pictures/Symbols:** Each picture stands for a certain number of items.
- **Key:** A small box that explains what one picture represents. **This is very important!**
- Example: Key: 🍎 = 2 children.

Creating a Simple Pictogram: Let's say we ask 5 friends their favourite snack:

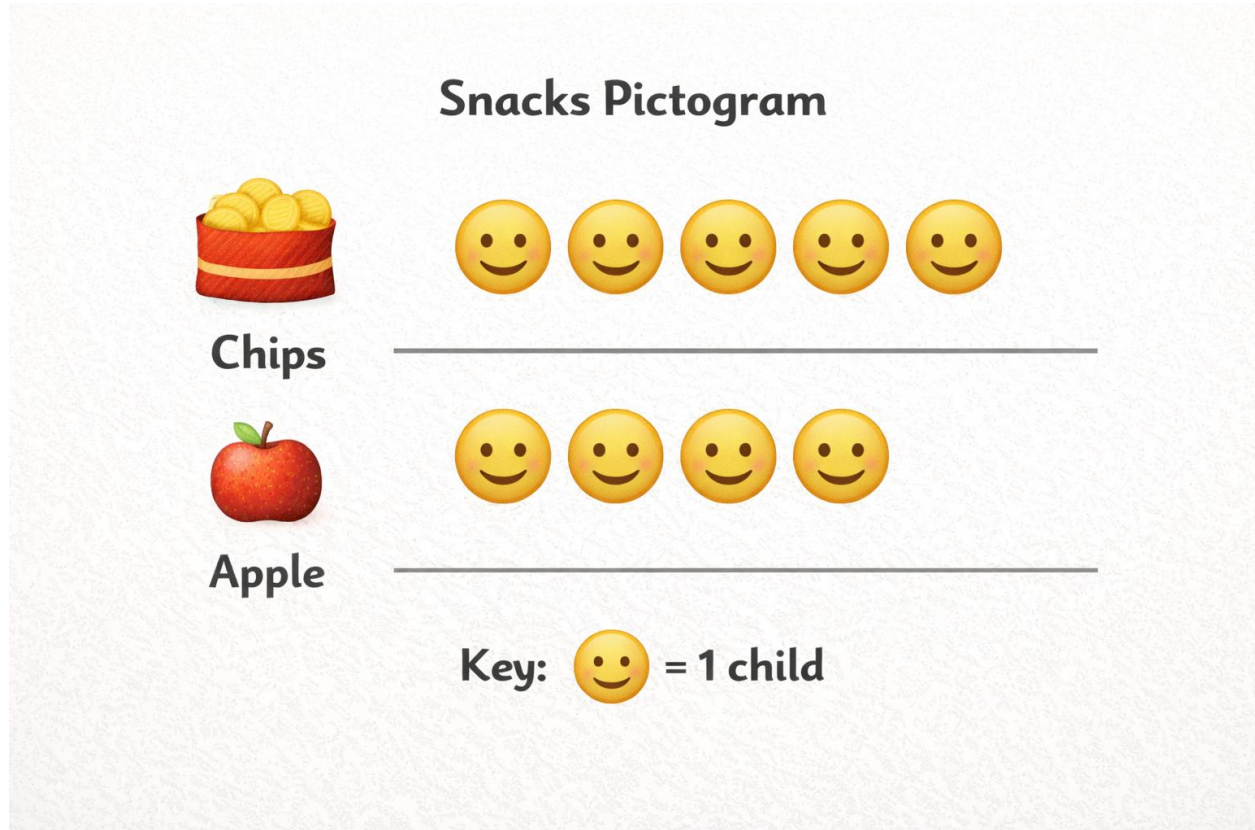
- Biscuit: 2 friends
- *Chin-Chin*: 5 friends
- *Kuli-Kuli*: 3 friends

We can draw it like this (if our Key is: 😊 = 1 friend): **Title:** Our Favourite Snacks

- Biscuit: 😊 😊
- *Chin-Chin*: 😊 😊 😊 😊 😊
- *Kuli-Kuli*: 😊 😊 😊

Interpreting a Pictogram: From our pictogram above:

1. Which snack is the **most** popular? (*Chin-Chin* has the most pictures).
2. Which snack is the **least** popular? (Biscuit has the fewest pictures).
3. How many friends like *Kuli-Kuli*? (Count the pictures: 3).



EVALUATION

1. Why do we use pictures in a pictogram instead of just writing numbers?
2. Look at this Key: ★ = 2 books. How many books do 3 stars represent?
3. What must every good pictogram have to explain the pictures?
4. In a survey, 4 children have dogs and 6 have cats. Draw a simple pictogram to show this.
(You can use □ for 1 child).
5. From your pictogram in Q4, how many more children have cats than dogs?

ASSIGNMENT

- **(Data Collection):** Ask 5 people in your family their favourite colour (Red, Blue, Green, Yellow). Tally the results.

- **(Drawing):** Use the data from Assignment 1 to draw a pictogram. Choose a simple symbol (like a ●) and let it stand for 1 person. Give your pictogram a title and a key.
 - **(Oral):** Show your pictogram to someone. Ask them two questions about the data, like "Which colour is most liked?" and "How many people like Green?".
-



WEEKS 12-13: REVISION AND EXAMINATION.

OBJECTIVES By the end of the lesson, pupils should be able to:

- Recall and solve questions from all topics covered in the term.

Comprehensive Revision Guide: We will revise all Third Term topics. Remember the big ideas:

- **Open Sentences:** Finding the missing number. Use the opposite operation.
- **Money:** Know your notes and coins. Add costs, subtract to find change.
- **Length:** Non-standard (hand span) vs. Standard (cm, m). $100\text{ cm} = 1\text{ m}$.
- **Time:** Read analogue clocks (o'clock, half past, quarter past/to).
- **Weight:** Heavy vs. Light. Units: g and kg. $1000\text{ g} = 1\text{ kg}$.
- **Capacity:** Full vs. Empty. Units: ml and l. $1000\text{ ml} = 1\text{ l}$.
- **Symmetry:** The mirror-image fold. Finding lines of symmetry.
- **Pictograms:** Reading and making picture graphs. Always check the Key!

Examination Preparation:

- Review your notes and assignments.
- Practice drawing clocks and reading the time.
- Practice measuring items at home with a ruler.
- Stay calm and think carefully during the exam.

EVALUATION

- **Weeks 12:** Intensive revision sessions with practice questions and quizzes.
- **Week 13:** End-of-Term Examination covering all Third Term topics.

ASSIGNMENT

- **(Written):** Make a list of all the topics for this term. Put a ✓ next to the ones you understand well and a ? next to the ones you need to review.
- **(Oral):** Explain to a parent or guardian what "symmetry" means. Find a symmetrical object in your house to show them.

- **(Review):** Solve these two mixed problems: a) A pencil costs ₦15. A ruler costs ₦10. How much for both? If you pay ₦50, what is your change? b) Draw a clock showing half past 10.

